# UNIT 5

# ROBOTICS

- The subject of robotics covers many different areas. Robots alone are hardly ever useful. They are used together with other devices, peripherals, and manufacturing machines.
- They are generally integrated into a system, which as a whole is designed to perform a task or to do an operation.
- They are capable of performing many different tasks and operations precisely and do not require common safety and comfort elements that humans needs.
- However, it takes much effort and many resources to make a robot function properly. As with humans, robots can do certain things, but not other things. As long as they are designed properly for the intended purpose, they are very useful and will continue to be used.
- Robotics is the art, knowledge base, and the know-how of designing, applying, and using robots in human endeavors.

# HISTORY OF ROBOTS

- The word robot always refers to an automated multifunctional manipulator that works by energy, to perform a variety of tasks.
- The word robot was introduced in 1920 in a play by Karel Capek called R.U.R., or Rossum's Universal Robots.
- Robot comes from the Czech word robota, meaning forced labor or drudgery.
- One of the first types of robots was a feedback (self-correcting) control mechanism. It was a watering trough that used a float to sense the water level. When the water gets to low, the float drops, opens a valve, and more water dumps into the trough.
- As the water rises, so does the float. Once it reaches a certain height, the valve is closed and the water is shut off. In 1954, the American inventor George Devol, Jr. developed a primitive arm that could be programmed to perform specific tasks.

# HISTORY OF ROBOTS

- In 1975, the American mechanical engineer Victor Scheinman developed a truly flexible multipurpose manipulator known as the Programmable Universal Manipulation Arm (PUMA).
- PUMA was capable of moving an object and placing it with any orientation in a desired location. Before the 1960s, robot usually meant a manlike mechanical device (mechanical man or humanoid) capable of performing human tasks or behaving in a human manner.
- Today robots come in all shapes and sizes, including small robots and larger wheeled robots that play football with a full-size ball. In 1995, there were about 700,000 robots operating in the world. A major user of robots is the automobile industry. General Motors uses approximately 16,000 robots for tasks such as spot welding, painting, machine loading, parts transfer, and assembly

# DEFINITION OF A ROBOT

A robot is a computer-controlled machine that is programmed to move, manipulate objects, and accomplish work while interacting with its environment. According to the Robot Institute of America (1979), a robot is defined as “a reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through various programmed motions for the performance of a variety of tasks.”

Or

Industrial robot is defined as “a number of rigid links connected by joints of different types that are controlled and monitored by computer.”

# INDUSTRIAL ROBOT

Industrial robots are advanced automation systems, mainly controlled by a computer. Today computers form an important part of industrial automation.

They supervise production lines and control manufacturing systems (e.g., machine tools, welders, laser cutting devices, etc.).

The new generation of robots executes various tasks in industrial systems and they participate in the full automation of factories.

Japanese defined industrial robots in four levels:

- Manual manipulators: perform fixed or preset task sequences.
- Playbacks: repeat pre-programmed fixed instructions.
- NC robot: carry out tasks through numerically loaded information.
- Intelligent robots: perform through their own recognition capabilities.

# INDUSTRIAL ROBOT

- A robot and a crane are very similar in the way they operate and in the way they are designed. Both possess a number of links attached serially to each other with joints, where some type of actuator can move each joint. In both systems, the hand of the manipulator can be moved in space and be placed in any desired location within the workspace of the system, each one can carry a certain amount of load and each one is controlled by a central controller which controls the actuators. However, one is called a robot and the other manipulator (or, in this case, a crane).
- The fundamental difference between the two is that in the case of a crane, a human operates and controls the actuators, whereas a computer that runs a program controls the robot manipulator.

# LAWS OF ROBOTICS

Isaac Asimov proposed three laws of robotics and he later added a "zeroth law."

- Law One: A robot may not injure a human being, or, through inaction, allow a human being to come to harm, unless this would violate a higher order law.
- Law Two: A robot must obey orders given to it by human beings, except where such orders would conflict with a higher order law.
- Law Three: A robot must protect its own existence as long as such protection does not conflict with a higher order law.
- Law Zero: A robot may not injure humanity, or, through inaction, allow humanity to come to harm.

# MOTIVATING FACTORS

For robotics systems to be introduced to the industrial world, they must have positive factors that would make a difference in using them. The motivating factors can be categorized as:

- Technical factors
- Economic factors
- Social factors

# 1.Technical Factors

Robots can do incomparable tasks that humans can’t do. It is generally considered that humans can’t match the speed, quality, reliability, and the endurance of a robotic system. In that they offer:

- High flexibility of product type and variation.
- Lower preparation time than hard automation.
- Better quality of products.
- Fewer rejects and less waste than labor intensive production

# 2. Economic Factors

- The needs to increase production rates to remain competitive.
- Pressure from the market place to improve quality.
- Increasing costs. Shortage of skilled labor.

# 3. Social Factors

- Some people think that the use of robotized systems increases the unemployment of workers and prevents many people from a main income. But the usage of robots causes a reduction of workload on workers and prevents dangerous working conditions as robots can be used in hazardous environments.

# ADVANTAGES AND DISADVANTAGES OF ROBOTS

- Robots offer specific benefits to workers, industries, and countries. If introduced correctly, industrial robots can improve the quality of life by freeing workers from dirty, boring, dangerous, and heavy labor.
- It can be said, therefore, that robots give the possibility to humans to occupy with jobs, that they can execute better.
- It is true that robots can cause unemployment by replacing human workers but robots also create jobs such as robot technicians, salesmen, engineers, programmers, and supervisors.

# Advantages of Robots

- Increase in productivity, safety, efficiency, quality, and consistency of products with the use of robots.
- Robots can work in hazardous environments without the need for life support, comfort, or concern about safety.
- Robots need no environmental comfort such as lighting, air conditioning, ventilation, and noise protection.
- Robots can work continuously without experiencing fatigue or boredom, do not have hangovers, and need no medical insurance or vacation.
- Robots have repeatable precision at all times, unless something happens to them or unless they wear out.
- Robots can be much more accurate than human

# Disadvantages of Robots

- Robots replace human workers creating economic problems, such as lost salaries, and social problems such as dissatisfaction and resentment among workers.
- Robots lack capability to respond in emergencies, unless the situation is predicted and the response is included in the system.
- Safety measures are needed to ensure that they do not injure operators and machines working with them.
- Robots are costly due to initial cost of equipment, installation costs, need for training, and need for programming

# CHARACTERISTICS OF AN INDUSTRIAL ROBOT

- An industrial robot has a hand, wrist, arm, base, lifting power, repeatability, manual control, automatic control, memory, programs, safety interlock, speed of operation, computer interface, reliability, and easy maintenance.
- The hand of a robot is known as a gripper or end effector or end-of-arm tooling. It is the driven mechanical device(s) attached to the end of the manipulator.
- The wrist of the robot is used to aim the hand at any part of the work piece. The wrist may have three motions: pitch (up-and-down motion), yaw (side-to-side motion), and roll (rotating motion).

# CHARACTERISTICS OF AN INDUSTRIAL ROBOT

- The arm is used to move the hand within reach of a part or work piece. It can pivot at its elbow and at its shoulder joint.
- The waist, or base, of the robot supports the arm and is called the shoulder. The arm can rotate about the shoulder.
- Repeatability is the replication of motion within some specified precision or tolerance.
- An RCC or remote center compliance device helps pull the hand or tool into the required position by acting as a multi-axis float.
- A manual control device is used to teach the robot how to do a new task.

# CHARACTERISTICS OF AN INDUSTRIAL ROBOT

- An automatic control system is used to carry out the instructions stored in the robot's memory.
- The robot's memory holds a library of programs to use in executing different tasks.
- Safety interlocks prevent the robot from inserting a hand into a machine and causing damage to the robot and machine.
- The robot's speed of operation in performing a task should be at least equal to that of the human worker it is replacing.
- The robot's computer interface enables the robot to use the computer's larger memory to hold more task programs and to synchronize its actions with a complete production line of robots and other machines.

# COMPONENTS OF AN INDUSTRIAL ROBOT

Industrial robot systems consist of four major subsystems as shown in Figure 13.1. Mechanical unit Drive Control system Tooling

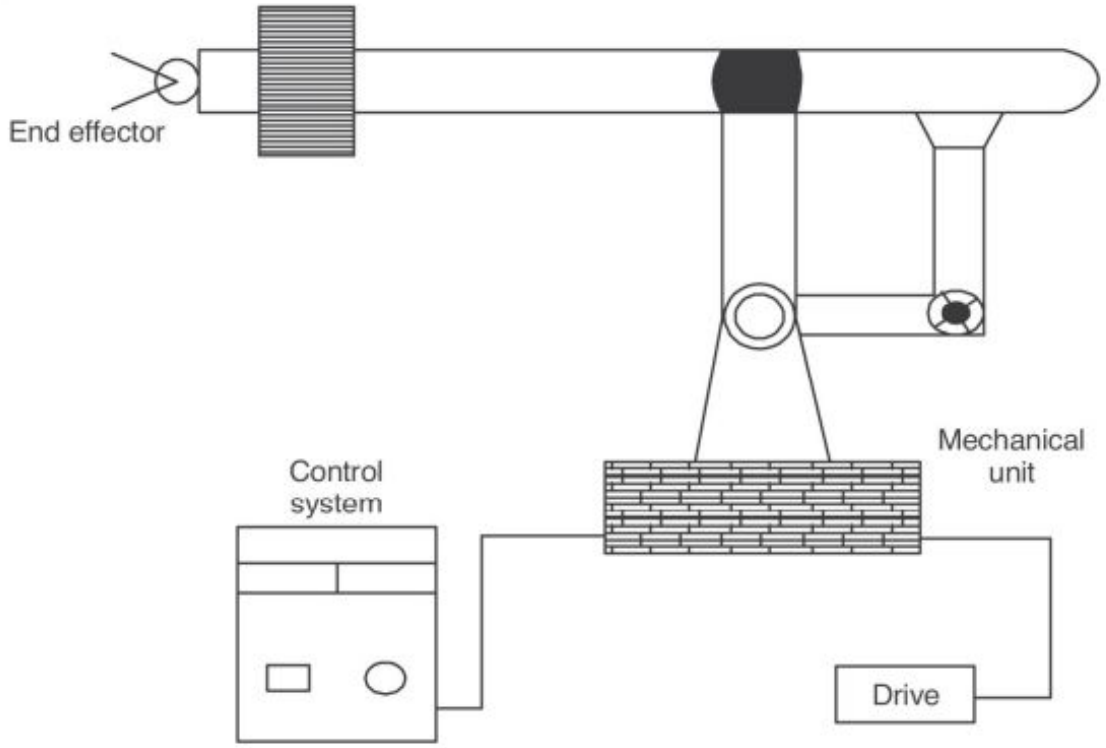



*FIGURE 13.1* Components of an Industrial Robot.

# COMPONENTS OF AN INDUSTRIAL ROBOT

- Mechanical Unit. The mechanical unit refers to the robot's manipulative arm and its base. Tooling such as end effectors, tool changers, and grippers are attached to the wrist-tooling interface. The mechanical unit consists of a fabricated structural frame with provisions for supporting mechanical linkage and joints, guides, actuators, control valves, limiting devices, and sensors. The physical dimensions, design, and loading capability of the robot depends upon the application requirements.

- Drive. An important component of the robot is the drive system. The drive system supplies the power, which enables the robot to move. Drive for a robot may be hydraulic, pneumatic, or electric. Hydraulic drives have been used for heavier lift systems. Pneumatic drives have been used for high speed, non-servo robots and are often used for powering tooling such as grippers. Electric drive systems can provide both lift and/or precision, depending on the motor and servo system selection and design. An AC or DC powered motor may be used depending on the system design and applications.

# COMPONENTS OF AN INDUSTRIAL ROBOT

- Control System. Controller is the brain of the robot. Controller is a communication and information-processing device that initiates, terminates, and coordinates the motions and sequences of a robot. Most industrial robots incorporate computer or microprocessor based controllers. These perform computational functions and interface with sensors, grippers, tooling, and other peripheral equipment.
- Controller programming may be done on-line or from off-line control stations. Programs may be on cassettes, floppy disks, internal drives, or in memory; and may be loaded or downloaded by cassettes, disks, or telephone modem. Some robot controllers have sufficient computational ability, memory capacity, and input/output capability to serve as system controllers for other equipment and processes.

# COMPONENTS OF AN INDUSTRIAL ROBOT

- Tooling. Tooling is manipulated by the robot to perform the functions required for the application. Depending on the application, the robot may have one functional capability, such as making spot welds or spray-painting.
- These capabilities may be integrated with the robot's mechanical system or may be attached at the robot's wrist-end effector interface. Alternatively, the robot may use multiple tools that may be changed manually (as part of set-up for a new program) or automatically during a work cycle.
- Tooling and objects that may be carried by a robot's gripper can significantly increase the envelope in which objects or humans may be struck.
- Tooling manipulated by the industrial robot and carried objects can cause more significant hazards than motion of the bare robotic system.
- The hazards added by the tooling should be addressed as part of the risk assessment.

# COMPONENTS OF AN INDUSTRIAL ROBOT

- Sensors. Sensors are used to collect information about the internal state of the robot or to communicate with the outside environment. As in humans, the robot controller needs to know where each link of the robot is, in order to know the robot's configuration.
- The state of the human body is determined because feedback sensors in human's central nervous system embedded in their muscle tendons send information to the brain.
- The brain uses this information to determine the length of their muscles, and thus, the state of their arms, legs, etc. The same is true for robots; sensors integrated into the robot send information about each joint or link to the controller, which determines the configuration of the robot.
- Robots are often equipped with external sensory devices such as a vision system, touch and tactile sensors, speech synthesizers, etc., which enable the robot to communicate with the outside world.

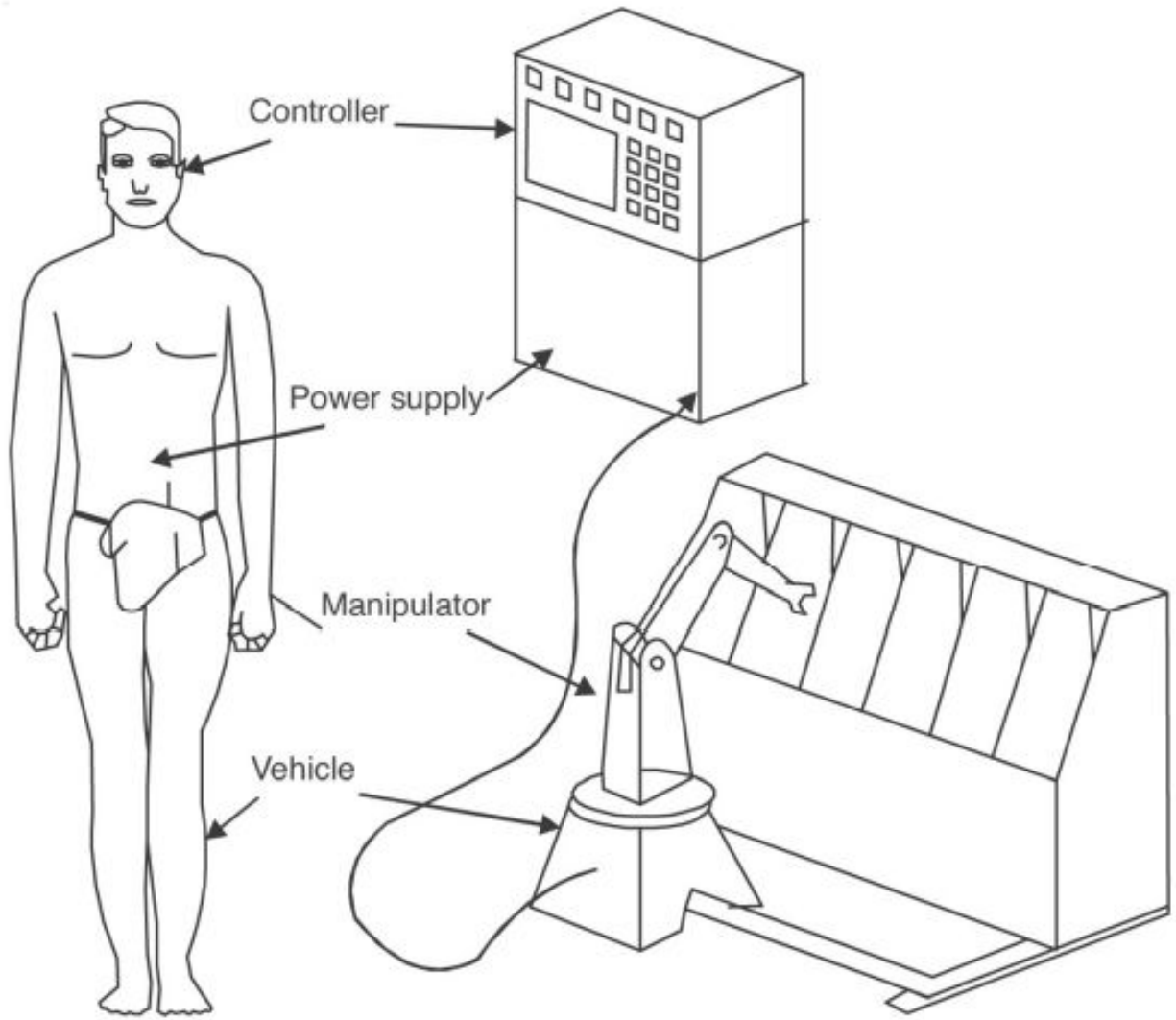
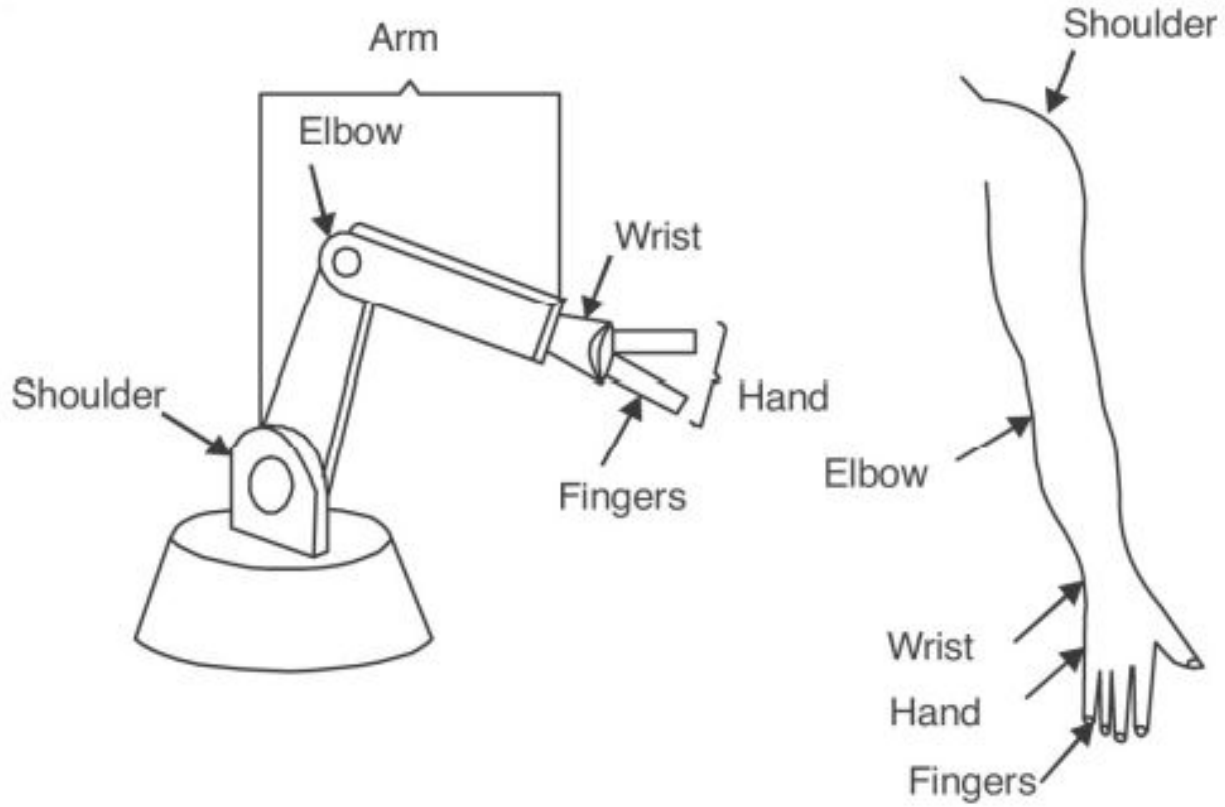



***FIGURE 13.2*** Comparison of Parts of Robot and Human Being.

# COMPARISON OF THE HUMAN AND ROBOT MANIPULATOR

- The parts of a robot's manipulator are named after similar parts in a human's chief manipulators, i.e., the arm and the hand.



# ROBOT WRIST AND END OF ARM TOOLS

- Six axis coordinates are required by a robot in order to completely specify the location and orientation of an object. Three coordinates can locate the center of gravity of an object (e.g., x, y, and z coordinates-in a rectangular coordinate system).
- Three other coordinates are normally achieved by adding wrist and hand movements with the end of arm tooling.
- There are three basic types of wrist motions:
  - Pitch - Rotational or bending movement in a vertical plane.
  - Yaw - Rotational or twisting movement in a horizontal plane.
  - Roll - Rotational or swivel movement.

# ROBOT WRIST AND END OF ARM TOOLS

- Additionally, the wrist (refer to Figure 13.4) serves as the mounting point for a variety of devices. These end effectors are either hand, or gripping devices, or job specific tool

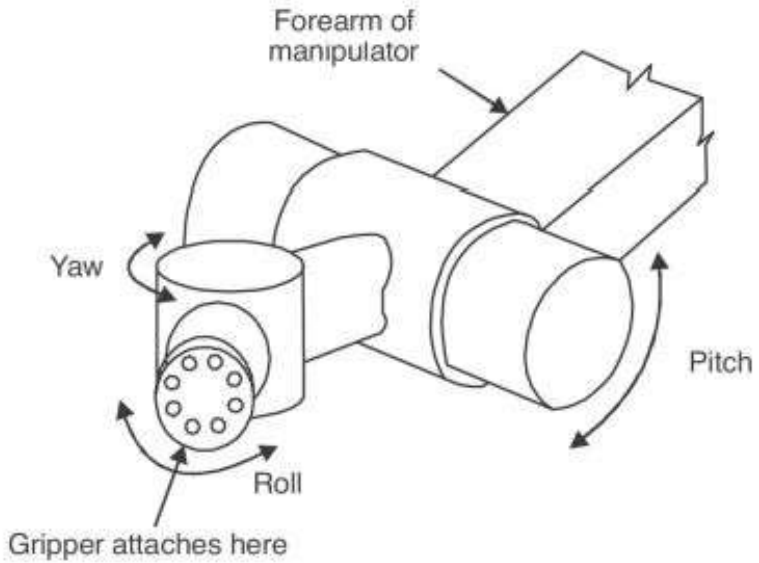



***FIGURE 13.4*** Robotic Wrist.

# ROBOT WRIST AND END OF ARM TOOLS

- Figure 13.5 shows the various arms and wrist motions of a robot.

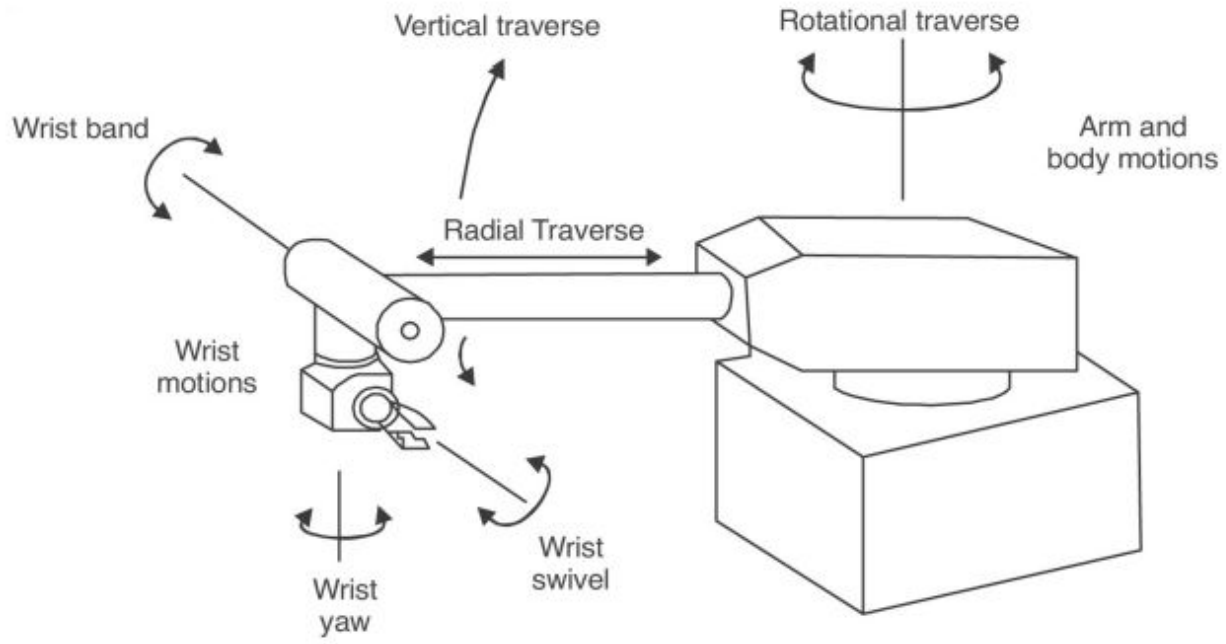



*FIGURE 13.5* Arm and Wrist Motions of a Robot.

# ROBOT WRIST AND END OF ARM TOOLS

- End Effector. The end effector is the device at the end of the robot arm. The end effector attached to the robot wrist is shown in Figure 13.6.

There are two main types of end effectors: grippers and tools.

i. Grippers: Grippers are devices, which can be used for holding or gripping an object. These include mechanical hands and anything like hooks, magnets, and suction devices, which can be used for holding or gripping.

ii. Tools: Tools are devices, which robots use to perform operations on an object, e.g., drills, paint sprays, grinders, welding torches, and any other tool which get a specific job done.

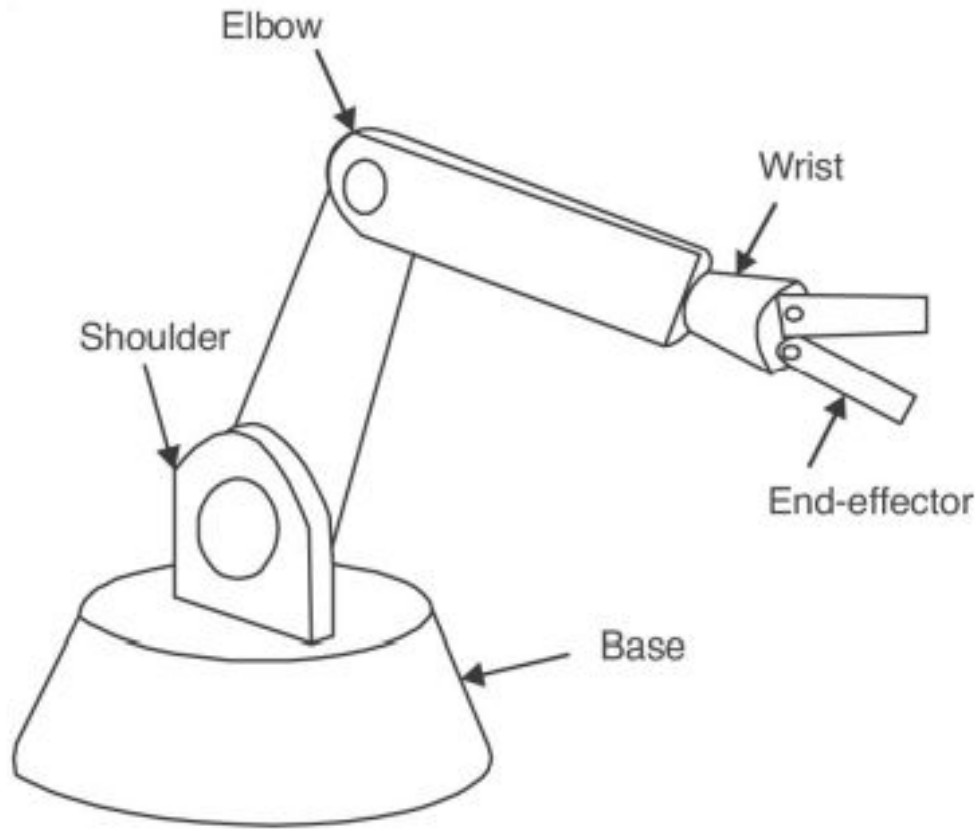



***FIGURE 13.6*** End Effector Attached to Robot Wrist.

# ROBOT TERMINOLOGY

There is a set of basic terminology and concepts common to all robots. These terms follow with brief explanations of each.

ROBOT TERMINOLOGY There is a set of basic terminology and concepts common to all robots. These terms follow with brief explanations of each.

i. Links and Joints: Links are the solid structural members of a robot, and joints are the movable couplings between them.

ii. Degree of Freedom (dof): Degree of freedom is the number of independent movements a robot can realize with respect to its base. The number of axes is normally the same as the number of degrees of freedom of the robot. Each joint on the robot introduces a degree of freedom. Each degree of freedom can be a slider, rotary, or other type of actuator. Robots typically have five or six degrees of freedom as shown in Figure 13.7.

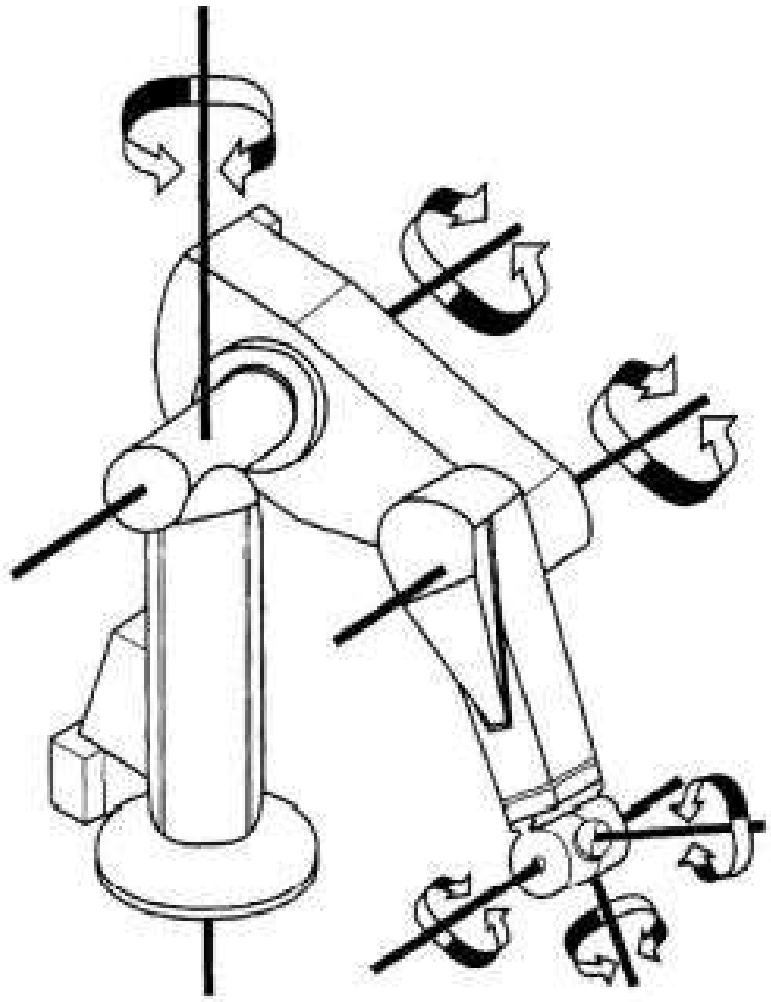


*FIGURE 13.7* Robot with Six Degrees of Freedom.

- Figure 13.8 shows an industrial robot with three basic degrees of freedom plus three degrees of freedom in the wrist and a seventh in its ability to move back and forth along the floor.

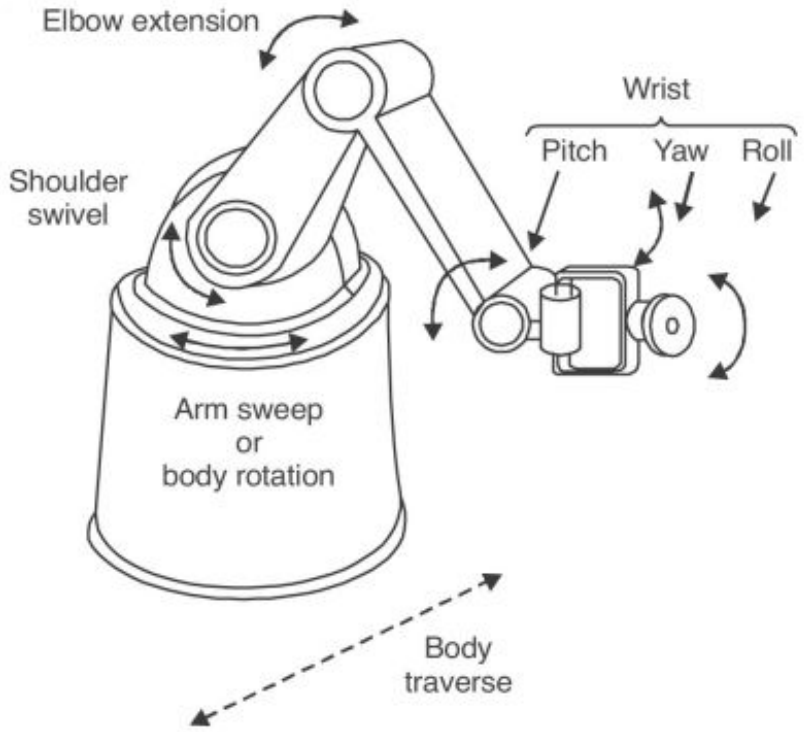
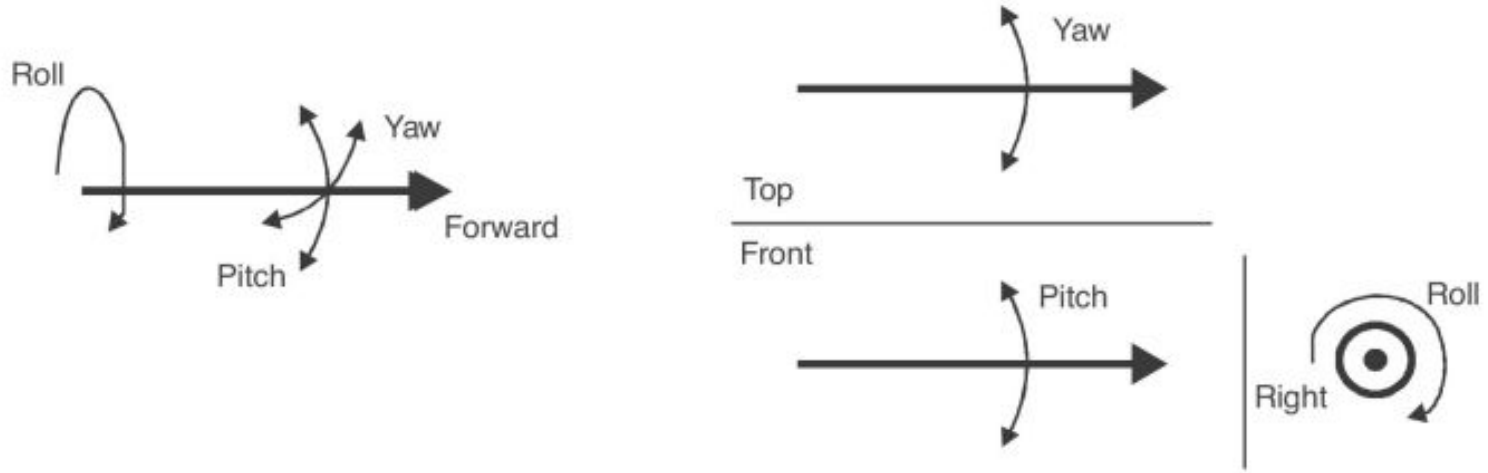



***FIGURE 13.8*** Robot with Seven Degrees of Freedom.

# ROBOT TERMINOLOGY

iii)Orientation Axis: Basically, if the tool is held at a fixed position, the orientation determines which direction it can be pointed in. Roll, pitch, and yaw are the common orientation axes used.



# ROBOT TERMINOLOGY

iv) Position Axis: The tool, regardless of orientation, can be moved to a number of positions in space.

v) Tool Center Point (TCP): The tool center point is located either on the robot, or the tool as shown in Figure 13.10. Typically, the TCP is used when referring to the robots position, as well as the focal point of the tool (e.g., the TCP could be at the tip of a welding torch). The TCP can be specified in Cartesian, cylindrical, spherical, etc., coordinates depending on the robot. As tools are changed, one often reprograms the robot for the TCP.

# ROBOT TERMINOLOGY

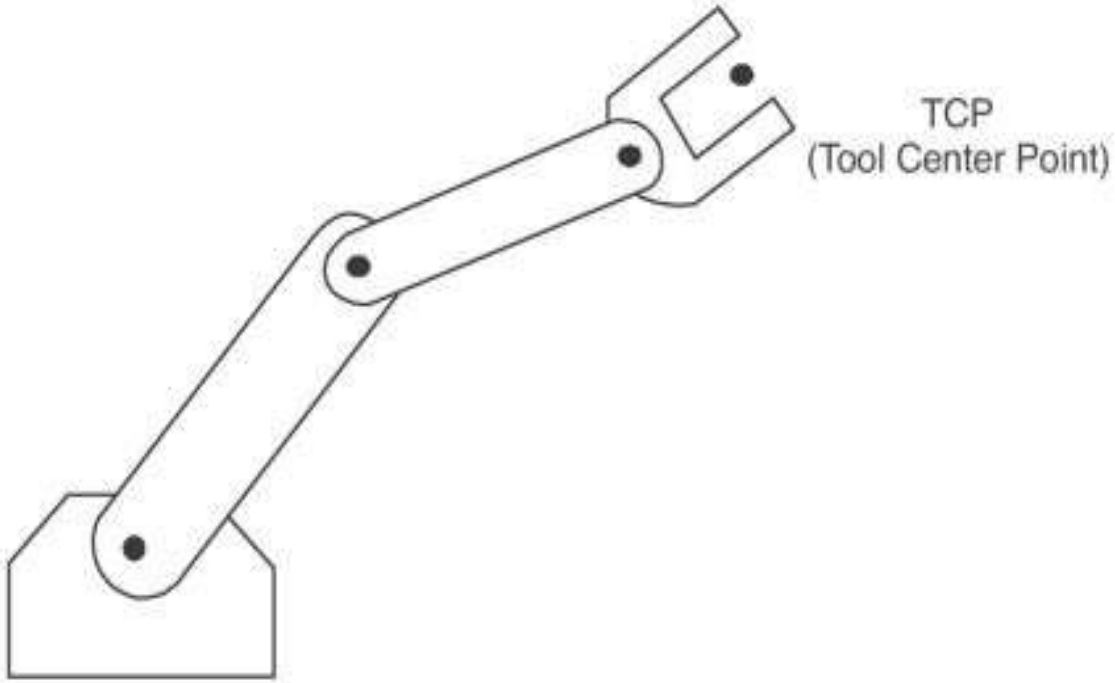



*FIGURE 13.10* Tool Center Point.

# ROBOT TERMINOLOGY

vi) Accuracy: Accuracy specification describes how close the arm will be when it moves to the desired point.

vii) Precision (validity): Precision is defined as how accurately a specified point can be reached. This is a function of the resolution of the actuators, as well as its feedback devices

viii)Repeatability (variability): Repeatability is how accurately the same position can be reached if the motion is repeated many times (Refer to Figure 13.11). Suppose that a robot is driven to the same point 100 times. Because many factors may affect the accuracy of the position, the robot may not reach the same point every time, but will be within a certain radius from the desired point.

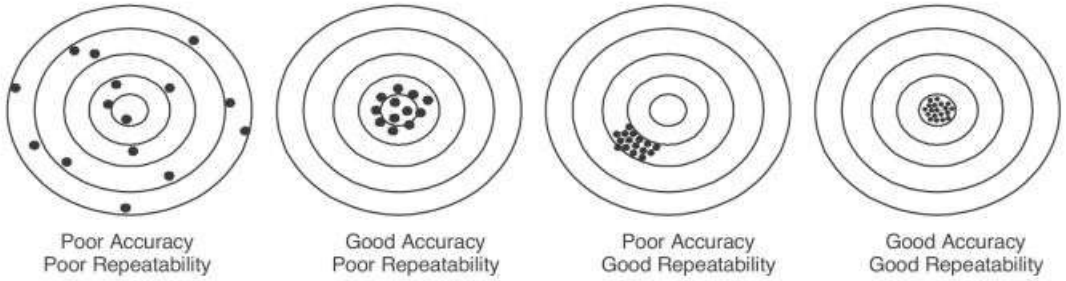



*FIGURE 13.11* Accuracy versus Repeatability.

# ROBOT TERMINOLOGY

xi) Work envelope/Workspace: A robot can only work in the area in which it can move. This area is called the work envelope. The work envelope is determined by how far the robot's arm can reach and how flexible the robot is. The more reach and flexibility a robot has, the larger the work envelope will be. It is one of the most important characteristics to be considered in selecting a suitable robot. Various robot configurations have different work envelopes. For a Cartesian configuration, the reach is a rectangular-type space.

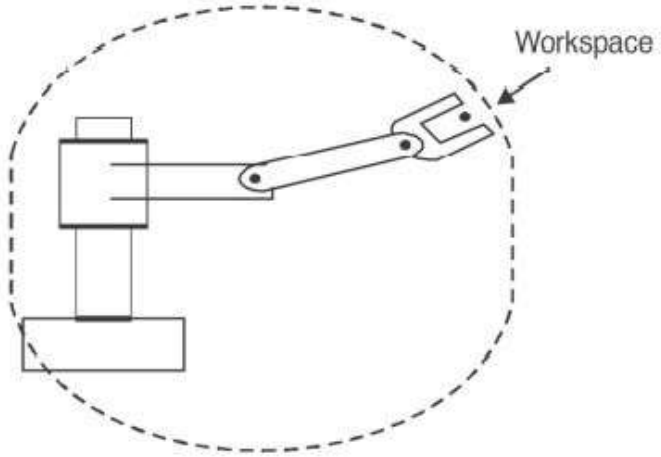



*FIGURE 13.12* Work Envelope of a Robot.

# ROBOT TERMINOLOGY

x) stability. Stability refers to robot motion with the least amount of oscillation. A good robot is one that is fast enough but at the same time has good stability.

xi) Speed. Speed refers either to the maximum velocity that is achievable by the Tool Center Point (TCP), or by individual joints. This number is not accurate in most robots, and will vary over the workspace as the geometry of the robot changes (and hence the dynamic effects). The number will often reflect the maximum safest speed possible. Some robots allow the maximum rated speed (100%) to be passed, but it should be done with great care.

# ROBOT TERMINOLOGY

xii) Payload. Payload is the weight a robot can carry and still remain within its specifications. For example, a robot's maximum load capacity may be much larger than its specified payload, but at the maximum level, it may become less accurate, may not follow its intended path accurately, or may have excessive deflections. The payload of robots compared with their own weight is usually very small.

xiii)Reach. Reach is the maximum distance a robot can reach within its work envelope.

xiv) Settling Time. During a movement, the robot moves fast, but as the robot approaches the final position it slows down, and slowly approaches at final position. The settling time is the time required for the robot to be within a given distance from the final position.

# ROBOTIC JOINTS

A robot joint is a mechanism that permits relative movement between parts of a robot arm. The joints of a robot are designed to enable the robot to move its end-effector along a path from one position to another as desired. The basic movements required for a desired motion of most industrial robots are:

- Rotational movement: This enables the robot to place its arm in any direction on a horizontal plane.
- Radial movement: This enables the robot to move its end-effector radially to reach distant points.
- Vertical movement: This enables the robot to take its end-effector to different heights.

# ROBOTIC JOINTS

Degrees of freedom, independently or in combination with others, define the complete motion of the end-effector. These motions are accomplished by movements of individual joints of the robot arm. The joint movements are basically the same as relative motion of adjoining links. Depending on the nature of this relative motion, the joints are classified as

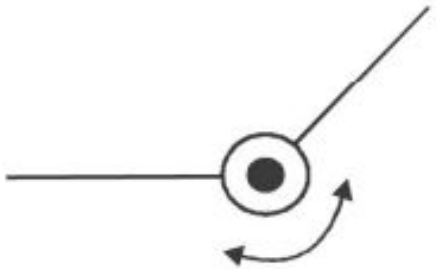
- Prismatic joints
- Revolute joints

# ROBOTIC JOINTS

- Prismatic joints are also known as sliding as well as linear joints. They are called prismatic because the cross section of the joint is considered as a generalized prism. They permit links to make a linear displacement along a fixed axis. In other words, one link slides on the other along a straight line. These joints are used in gantry, cylindrical, or similar joint configurations.
- Revolute joints, the second type of joint is a revolute joint where a pair of links rotates about a fixed axis.



(a) Prismatic joint



(b) Revolute joint

# ROBOTIC JOINTS

The variations of revolute joints shown in Figure 13.14 include: Rotational joint (R) Twisting joint (T) Revolving joint (V)

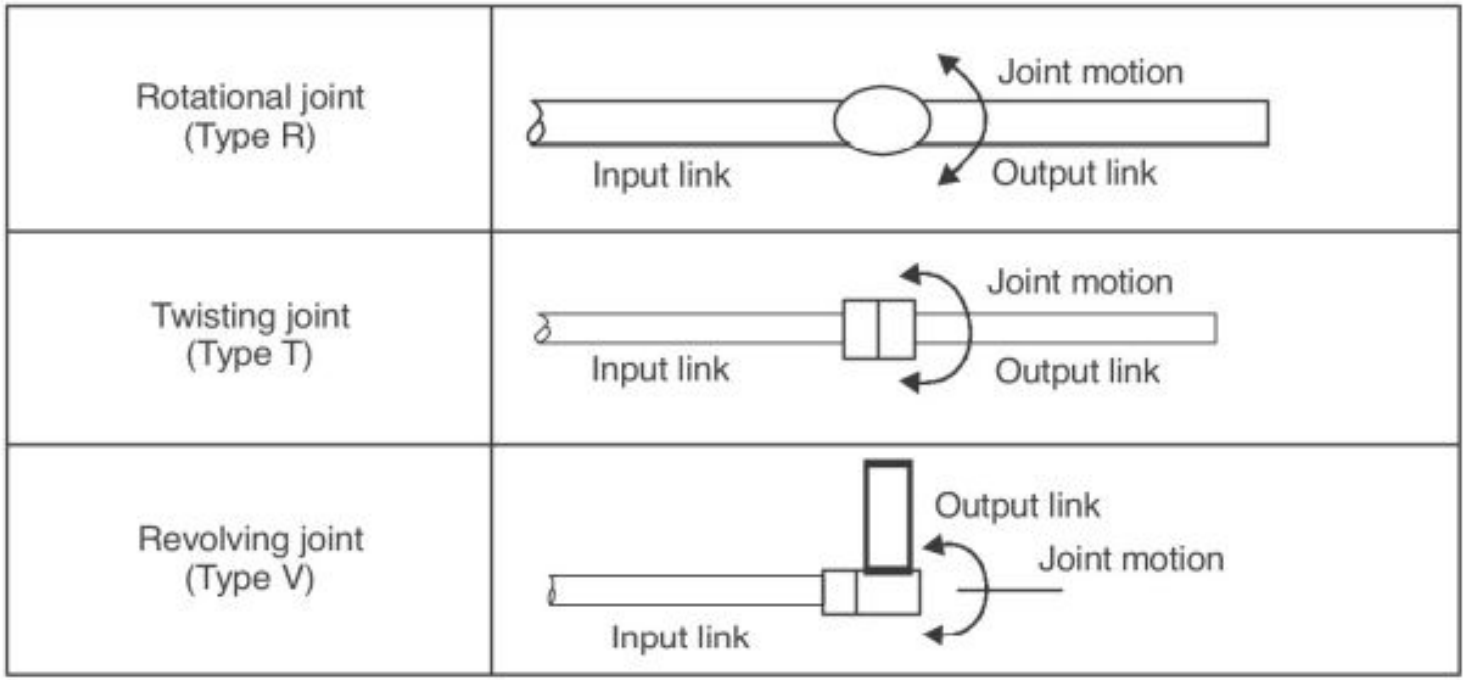



***FIGURE 13.14*** Revolute Joints.

# ROBOTIC JOINTS

- Rotational joint (R) is identified by its motion, rotation about an axis perpendicular to the adjoining links. Here, the lengths of adjoining links do not change but the relative position of the links with respect to one another changes as the rotation takes place.
- A twisting joint (T) is also a rotational joint, where the rotation takes place about an axis that is parallel to both adjoining links.
- A revolving joint (V) is another rotational joint, where the rotation takes place about an axis that is parallel to one of the adjoining links. Usually, the links are aligned perpendicular to one another at this kind of joint. The rotation involves revolution of one link about another.

# CLASSIFICATION OF ROBOTS

Robots can be classified by four fundamental elements of operation:
Coordinate systems

- Power source
- Method of control
- Programming method

# ROBOT CLASSIFICATION ON THE BASIS OF COORDINATE SYSTEMS

Structurally, robots are classified according to the wrist's coordinate system as follows

- Cartesian/Rectilinear Robot: The axis or dimensions of these robots are 3 intersecting straight lines (x-y-z) as shown in Figure 13.15. The Cartesian coordinate robot is one that consists of a column and an arm. It is sometimes called an x-y-z robot, indicating the axis of motion. The x-axis is lateral motion, the y-axis is longitudinal motion, and the z-axis is vertical motion. Thus, the arm can move up and down on the z-axis; the arm can slide along its base on the xaxis; and then it can telescope to move to and from the work area on the y-axis. The features of a Cartesian robot (electronic equipment, control program) are same as those of CNC machine tools

# Cartesian/Rectilinear Robot

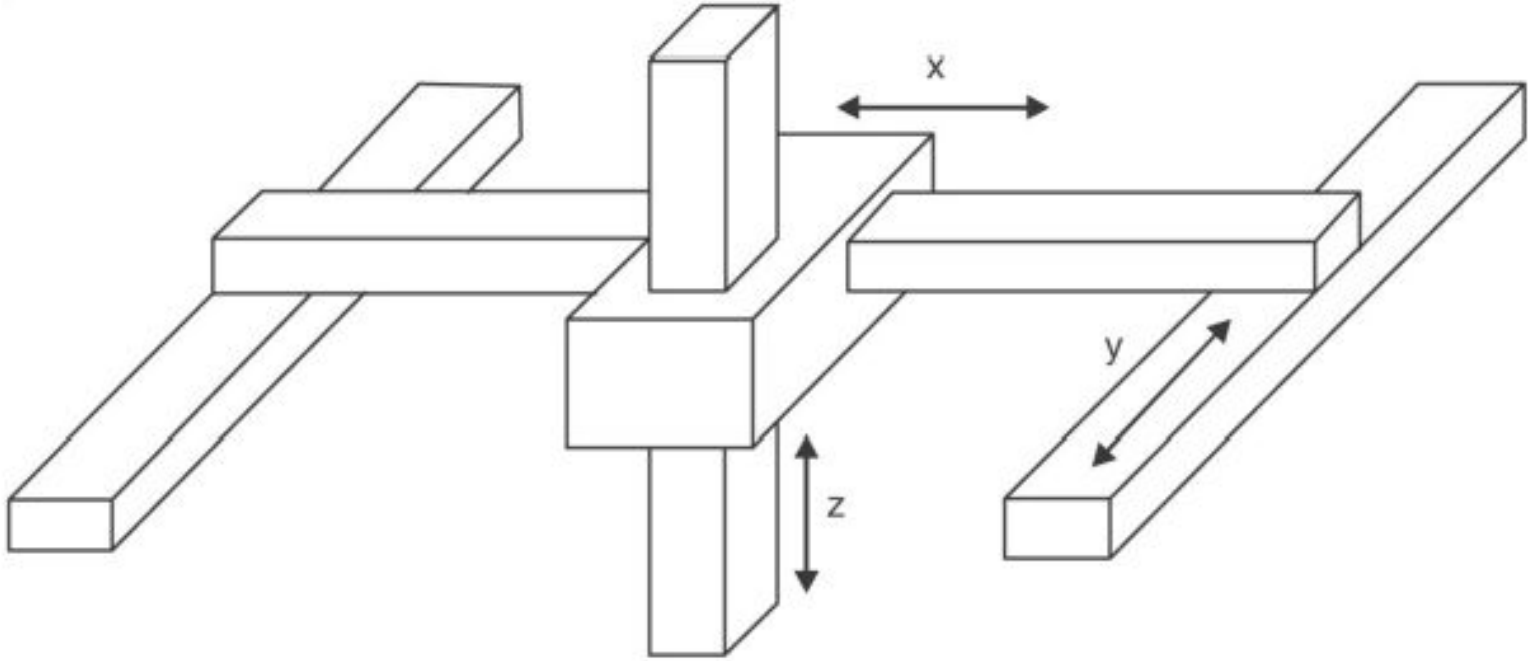



***FIGURE 13.15*** Cartesian Robot.

# Cartesian/Rectilinear Robot

Applications

- These types of robots are used for:
- Pick and place work
- Application of sealant
- Assembly operations Handling machine tools
- Arc welding

Advantages

- Ability to do straight line insertions into furnaces
- Easy computation and programming
- Most rigid structure for given length

Disadvantages

- Requires large operating volume
- Exposed guiding surfaces require covering in corrosive or dusty environments Can only reach front of itself
- Axis hard to seal

# Cylindrical Robot

- Cylindrical robots have one angular dimension and 2 linear dimensions as shown in Figure 13.16. The rigid structure of this system offers them the capability to lift heavy loads through a large working envelope. The main body of such a robot consists of a horizontal arm mounted on a vertical column. The column is mounted on a rotating base. The horizontal arm moves forward and backward on the direction of the longitudinal axis and it also moves up and down on the column. Column and arm are rotating on the base around the vertical axis.

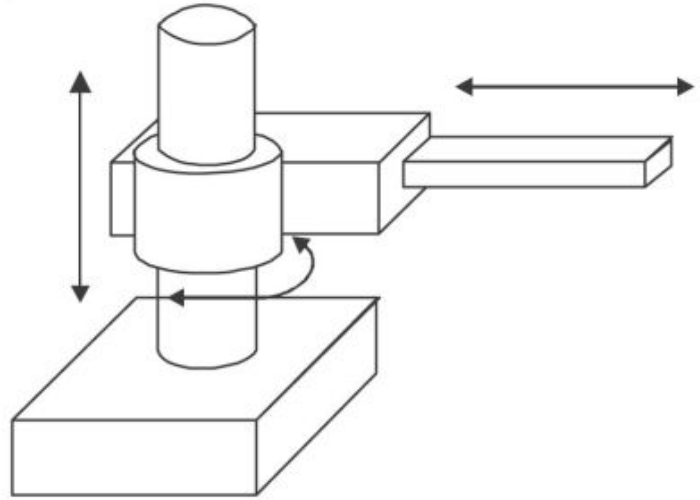


*FIGURE 13.16* Cylindrical Robot.

# Cylindrical Robot

Applications

- Assembly operations
- Handling machine tools
- Spot-welding
- Handling die-casting machines

Advantages

- Can reach all around itself
- Rotational axis easy to seal
- Relatively easy programming
- Rigid enough to handle heavy loads through large working space
- Good access into cavities and machine openings

Disadvantages

- Can’t reach above itself
- Won’t reach around obstacles
- Exposed drives are difficult to cover from dust and liquids Linear axes is hard to seal

# Spherical (Polar) Robot

Robots of this type consist of a rotating base, a lifting part, and a telescopic arm, which moves inwards and outwards (refer to Figure 13.17). The 2 dimensions of spherical robots are angles and the third is a linear distance from the point of origin. These robots operate according to spherical coordinates and offer greater flexibility. This design is used where a small number of vertical actions are adequate

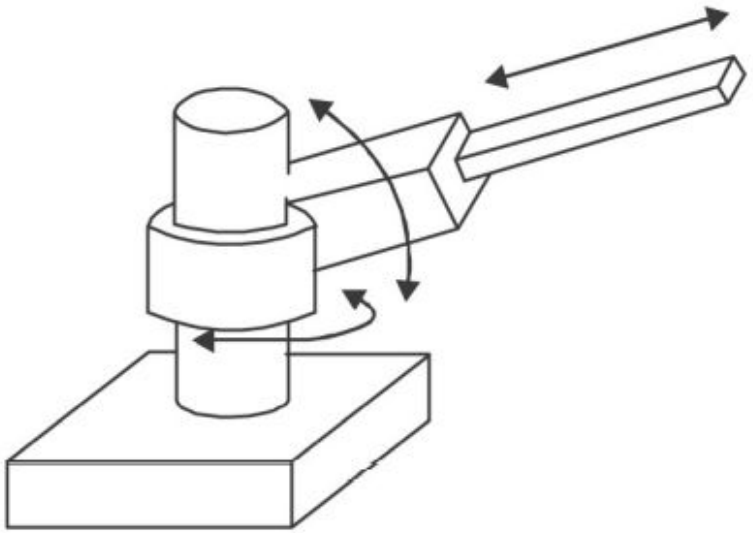


*FIGURE 13.17* Spherical Robot.

# Spherical (Polar) Robot

Applications

- Handlings at die casting or fettling machines
- Handling machine tools
- Arc/spot welding

Advantages

- Large working envelope
- Two rotary drives are easily sealed against liquids/dust

Disadvantages

- Complex coordinates more difficult to visualize, control, and program
- Exposed linear drive
- Low accuracy

# Articulated Robot

Articulated robots consist of three constant parts (links) that are joined with revolute joints and are placed on a rotating base as shown in Figure 13.18. The kinematics layout is similar to a human arm. The tool (gripper) is similar to a palm and is adjusted to the lower part of the arm through the wrist. The elbow connects the lower and upper part of the arm and the shoulder connects the upper part of the arm with the base. Many times the shoulder joint has a rotational motion in the horizontal plane. The articulated robot has all three axes revolute, so the position resolution is completely dependent on the arm's position. The total accuracy of an articulated robot is small, because joint errors are accumulated at the end of the arm, which is at the wrist position.

# Articulated Robot

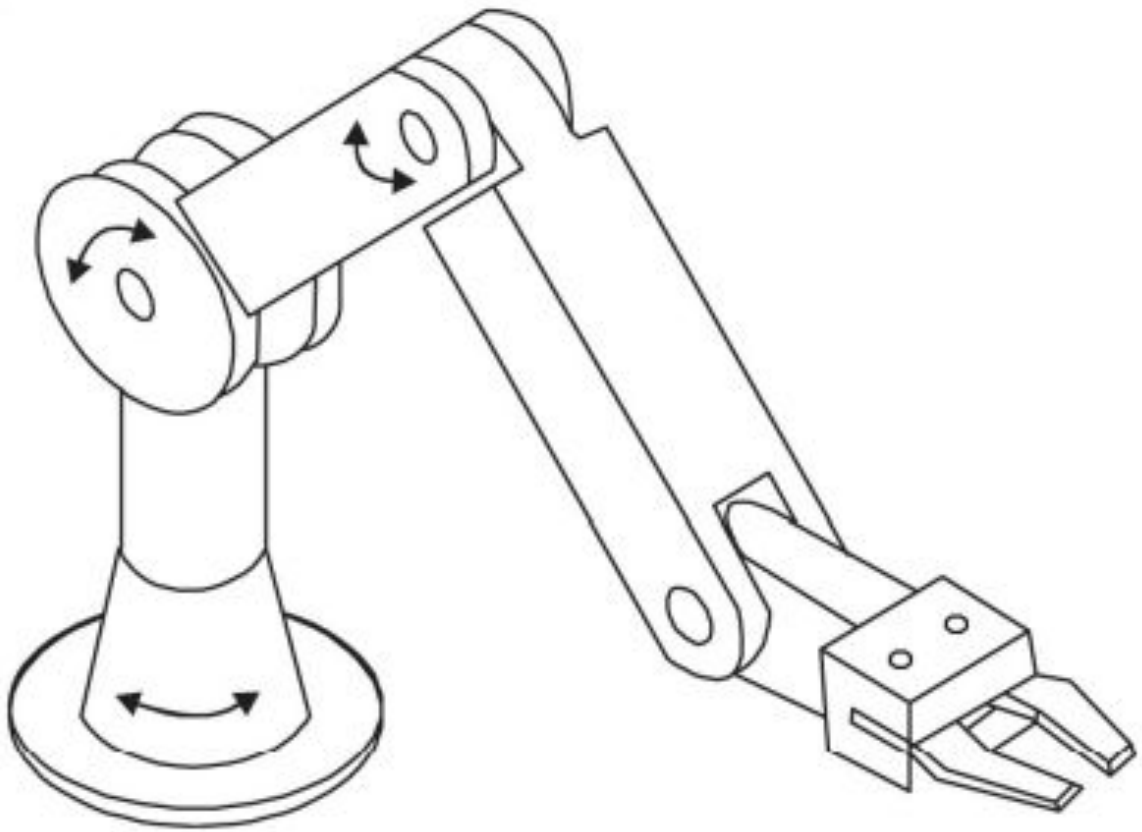


***FIGURE 13.18*** Articulated Robot.

# Articulated Robot

Applications

- Assembly operations
- Die casting
- Fettling machines
- Gas welding
- Arc welding
- Spray-painting

Advantages

- High mechanical flexibility
- They can move with high speed at three degrees of freedom
- All joints can be sealed from the environment

Disadvantages

- Extremely difficult to visualize, control, and program
- Restricted volume coverage
- Low accuracy

# SCARA Robot

One style of robot that has recently become quite popular is a combination of the articulated arm and the cylindrical robot. This robot has more than three axes and is called a SCARA robot (refer to Figure 13.19). It is used widely in electronic assembly. The rotary axes are mounted vertically rather than horizontally. This configuration minimizes the robot's deflection when it carries an object while moving at a programmed speed. The acronym SCARA stands for Selective Compliance Assembly Robot Arm, a particular design developed in the late 1970s.

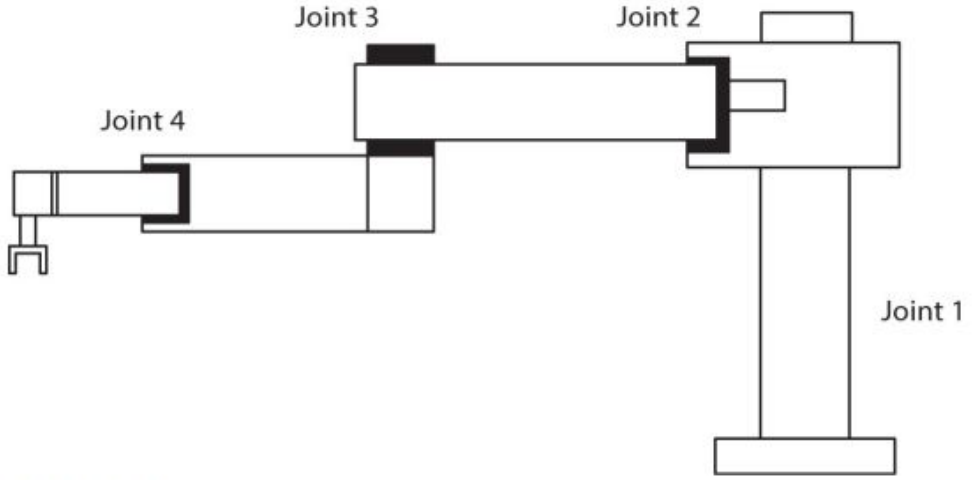



*FIGURE 13.19* SCARA Robot.

# SCARA Robot

Applications

- Pick and place work
- Assembly operations
- Application of sealant
- Handling machine tools

Advantages

- High speed Excellent repeatability
- Good payload capacity
- Large work area for floor space
- Moderately easy to program

Disadvantages

- Limited application
- Two ways to reach point
- Difficult to program off-line
- Highly complex arm

# ROBOT CLASSIFICATION ON THE BASIS OF POWER SOURCE

Actuators drive the mechanical linkages and joints of a manipulator, which can be various types of motors and valves. The energy for these actuators is provided by some power source such as hydraulic, pneumatic, or electric. There are three major types of drive systems for industrial robots:

- Hydraulic drive system
- Pneumatic drive system
- Electric drive system

# Hydraulic Drive System

- The most popular form of the drive system is the hydraulic drive system because hydraulic cylinders and motors are compact and allow for high levels of force and power, together with accurate control. These systems are driven by a fluid that is pumped through motors, cylinders, or other hydraulic actuator mechanisms. A hydraulic actuator converts forces from high pressure hydraulic fluid into the mechanical shaft rotation or linear motion. Hydraulic robots are preferred in environments in which the use of electric drive robots may cause fire hazards, for example, in spray painting.

Advantages

- A hydraulic device can produce an enormous range of forces without the need for gears, simply by controlling the flow of fluid Preferred for moving heavy parts
- Preferred to be used in explosive environments Self-lubrication and selfcooling
- Smooth operation at low speeds
- There is need for return line

Disadvantages

- Occupy large space area
- There is a danger of oil leak to the shop floor

# Pneumatic Drive System

- Pneumatic drive systems are found in approximately 30 percent of today's robots. These systems use compressed air to power the robots. Because machine shops typically have compressed air lines in their working areas, the pneumatically driven robots are very popular. These robots generally have fewer axis of movement and can carry out simple pick-andplace material-handling operations, such as picking up an object at one location and placing it at another location. These operations are generally simple and have short cycle times. The pneumatic power can be used for sliding or rotational joints.

Advantages

- Less expensive than electric or hydraulic robots Suitable for relatively less degrees of freedom design
- Do not pollute work area with oils
- No return line required Pneumatic devices are faster to respond as compared to a hydraulic system as air is lighter than fluid

Disadvantages

- Compressibility of air limits control and accuracy aspects
- Noise pollution from exhausts Leakage of air can be of concern

# Electric Drive System

Electrical drive systems are used in about 20 percent of today's robots. These systems are servomotors, stepping motors, and pulse motors. These motors convert electrical energy into mechanical energy to power the robot. Compared with a hydraulic system, an electric system provides a robot with less speed and strength. Electric drive systems are adopted for smaller robots. Electrically driven robots are the most commonly available and used industrial robots.

There are three major types of electric drive that have been used for robots:

- Stepper Motors: These are used mainly for simple pick and place mechanisms where cost is more important than power or controllability.
- DC Servos: For the early electric robots, the DC servo drive was used extensively. It gave good power output with a high degree of control of (c) both speed and position.
- AC Servos: In recent years, the AC servo has taken over from the DC servo as the standard drive. These modern motors give higher power output and are almost silent in operation. As they have no brushes, they are very reliable and require almost no maintenance in operation.

Advantages

- Good for small and medium size robots
- Better positioning accuracy and repeatability Less maintenance and reliability problems

Disadvantages

- Provides less speed and strength than hydraulic robots
- Not all electric motors are suited for use as actuators in robots
- Require more sophisticated electronic controls and can fail in high temperature, wet, or dusty environments

# ROBOT CLASSIFICATION ON THE BASIS OF METHOD OF CONTROL

- The motions of a robot are controlled by a combination of software and hardware that is programmed by the user. Robots are classified by control method into servo and non-servo robots.

# Non-Servo Controlled Robots

- Non-servo control is a purely mechanical system of stops and limit switches, which are pre-programmed for specific repetitive movements.
- This can provide accurate control for simple motions at low cost. The motions of non-servo controlled robots are controlled only at their endpoints, not throughout their paths. Non-servo robots are often referred as “endpoint,” “pick and place,” or “limited sequence” robots. These robots are used primarily for materials transfer.

Characteristics of non-servo robots include:

- Relatively high speed possible due to smaller size of the manipulator
- These robots are low cost and simple to operate, program, and maintain
- These robots have limited flexibility in terms of program capacity and positioning capability

# Servo Controlled Robots

Servo control system is capable of controlling the velocity, acceleration, and path of motion, from the beginning to the end of the path. It uses complex control programs. These systems use programmable logic controllers (PLCs) and sensors to control the motions of robots. They are more flexible than non-servo systems, and they can control complicated motions smoothly. Sensors are used in servo-control systems to track the position of each of the axis of motion of the manipulator. Servo controlled robots are classified according to the method that the controller uses to guide the end-effector. These are:

- Point-to-point (PTP)
- control robot Continuous-path (CP)
- control robot Controlled-path robot

# Point-to-Point Control Robot (PTP)

- A point-to-point robot is capable of moving from one discrete point to another within its working envelope.
- During point-to-point operation the robot moves to a position, which is numerically defined, and it stops there.
- The end effector performs the desired task, while the robot is halted. When task is completed, the robot moves to the next point and the cycle is repeated.

Applications:

- Component insertion
- Spot-welding
- Hole drilling
- Machine loading and unloading
- Assembly operations

# Continuous-path (CP) Control Robot

- In a continuous-path robot, the tool performs its task, while the robot (its axes) is in motion, like in the case of arc welding, where the welding pistol is driven along the programmed path. All axes of continuous path (c) robots move simultaneously, each with a different speed. The computer coordinates the speeds so that the required path is followed. The robot's path is controlled by storing a large number of spatial points in the robot's memory during the teach sequence

Applications:

- Spray painting
- Finishing
- Gluing Arc welding operations
- Cleaning of metal articles
- Complex assembly processes

# Controlled-path Robot

- In controlled-path robots, the robot is moved along a computer generated, predictable path as the robot travels from point to point.
- The computer-generated path may be a straight line with end-effector orientation or it may involve curved paths through successive points and/or gradual orientation changes.
- Good accuracy can be obtained at any point along the specified path. Only the start and finish points and the path definition function must be stored in the robot's control memory.
- The robot movements are more precise than with point-to-point programming and are less likely to present a hazard to personnel and equipment.

# ROBOT CLASSIFICATION ON THE BASIS OF PROGRAMMING METHOD

- Manual programming
- Lead-through programming
- Walk-through programming

# ROBOT SELECTION

Once the application is selected, a suitable robot should be chosen from the many commercial robots available in the market. The characteristics of robots generally considered in a selection process include:

- Size of class
- Degrees of freedom
- Velocity Drive type Control mode
- Repeatability
- Lift capacity
- Right-left,
- Up-down, and In-out traverse Yaw, pitch, and roll
- Weight of the robot

# ROBOT SELECTION

1. Size of class: The size of the robot is given by the maximum dimension (x) of the robot work envelope.

- Micro ($x < 1$ m) Small ($1$ m $< x < 2$ m)
- Medium ($2$ m $< x < 5$ m)
- Large ($x > 5$ m)

2. Degrees of freedom: The cost of the robot increases with the number of degrees of freedom. Six degrees of freedom is suitable for most works.

3. Velocity: Velocity consideration is effected by the robot's arm structure.

- Rectangular
- Cylindrical
- Spherical
- Articulated

# ROBOT SELECTION

4. Drive type:
- Hydraulic
- Electric
- Pneumatic

5. Control mode:
- Point-to-point control (PTP)
- Continuous path control (CP)
- Controlled path control

6. Lift capacity:
- 0–5 kg
- 5–20 kg
- 20–40 kg and so forth

# ROBOTIC SENSORS

- Sensors serve as a robot's sight, hearing, touch, taste, and smell. Without sensors, a robotic device would not be able to discern anything about current surroundings.
- A signal is returned from the sensor to the robot CPU and is applied to the current situation or saved for later analysis.
- Without sensors, a robot is just a machine. Sensors provide feedback to the control systems and give the robots more flexibility
- As robots are used in hazardous areas and industrial processes, they are fitted with sensors to gather information about the nearby environment and objects around, to come out with precise results of the work. R

# TYPES OF SENSORS IN ROBOTS

- The basic function of a sensor is to measure some feature of the world, such as light, sound, or pressure and convert that measurement into an electrical signal, usually a voltage or current.
- Typical sensors respond to stimuli by changing their resistance (photocells), changing their current flow (phototransistors), or changing their voltage output (the sharp IR sensor).
- The electrical output of a given sensor can easily be converted into other electrical representations. The control of the manipulator or industrial robot is based on correct interpretation of sensory information. Sensing includes activities such as seeing, hearing, touching, smelling, and measuring distance.
- They include not only transducers for physical quantities such as microphones for sounds, but also data processing input devices such as keyboard for textual information and specialized sensors.

# TYPES OF SENSORS IN ROBOTS

Sensors in robots can be classified as:

- Exteroceptors or External Sensors (for the measurement of robot's environmental parameters).
- Proprioceptors or Internal Sensors (for the measurement of robots internal parameters).

# EXTEROCEPTORS OR EXTERNAL SENSORS

Exteroceptors are sensors that measure the positional or force-type interaction of the robot with its environment. These sensors are added to robots to perceive the world in which they operate and interact with the environment outside the robot. External sensors can be categorized as:

- Contact sensors
- Non contact sensors

# Contact Sensors

Contact sensing is one of the most basic requirements for any manipulator interacting physically with the environment in a non-structured manner. It is the ability of the robot to determine the shape, size, weight, or even the surface texture of an object by touching it.

The functions of contact sensors in controlling manipulation may be classified into the following basic material handling and assembly operations:

- Searching—detecting a part by sensitive touch sensors on the hand exterior without moving the part.
- Recognition—determining the identity, position, and orientation of a part, again without moving it, by sensitive touch sensors with high spatial resolution.
- Grasping—acquiring the part by deformable, roundish fingers, with sensors mounted on their surfaces.
- Moving—placing, joining, or inserting a part with the aid of force sensors.

Most important types of robotic sensors of contact type are:

- Tactile sensors
- Force sensors

# Noncontact Sensors

These sensors are used to give the robot information about the process or the environment without the use of physical contact. Noncontact sensors include:

- Pneumatic sensors, which detect part presence by air, flow disturbance.
- Ultrasonic sensors that analyze sound waves reflected from a part.
- Proximity sensors that register the approach, arrival, or removal of parts.
- Optical sensors utilizing interrupted light beams across the path of an incoming part
- Machine vision systems that use visual sensors, usually video cameras, to provide data that allows the robot to make intelligent decisions regarding parts.

# TACTILE SENSORS

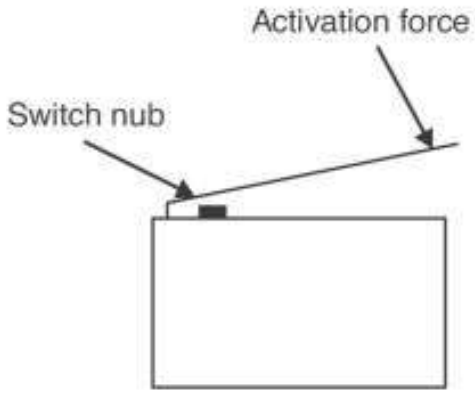
- Tactile sensors are used to identify and to control directly the interaction between the robot end effector and the environment. The interaction with the object may occur by direct contact.
- These sensors either detect when the hand touches something, or they measure some combination of force and torque components that the hand is exerting on an object.
- These sensors sense the external condition of an object, the roughness or smoothness of its surface, its slipperiness, its elasticity, etc. This sort of information is necessary in order to position the robot manipulators on an object and to compute the force and pressure that must be applied, say to grasp and lift the object.

Tactile sensors can be classified into:

- Touch sensors
- Stress sensors

# Touch Sensors

- Touch sensors produce a binary output signal, depending upon whether or not they are in contact with something. Touch sensors may be mounted on the outer and inner surfaces of each finger. The outer sensors may be used to search for an object and possibly determine its identity, position, and orientation. The inner mounted sensors may be used to obtain information about an object before it is acquired and about grasping forces and workpiece slippage during acquisition. Microswitchs are the most commonly used and least expensive kind of touch sensor which either turns on or off as contact is made. A microswitch is shown in Figure 14.1.



# Touch Sensors

- The bumper skirt on the robot shown in Figure 14.2 is an example of a touch sensor. When the robot runs into a wall, the bumper skirt hits a micro switch which lets the robot controller know that the robot is up against a wall. Other types of touch sensors are used internally to let the robot know when an arm is extended too far and it should be retracted.

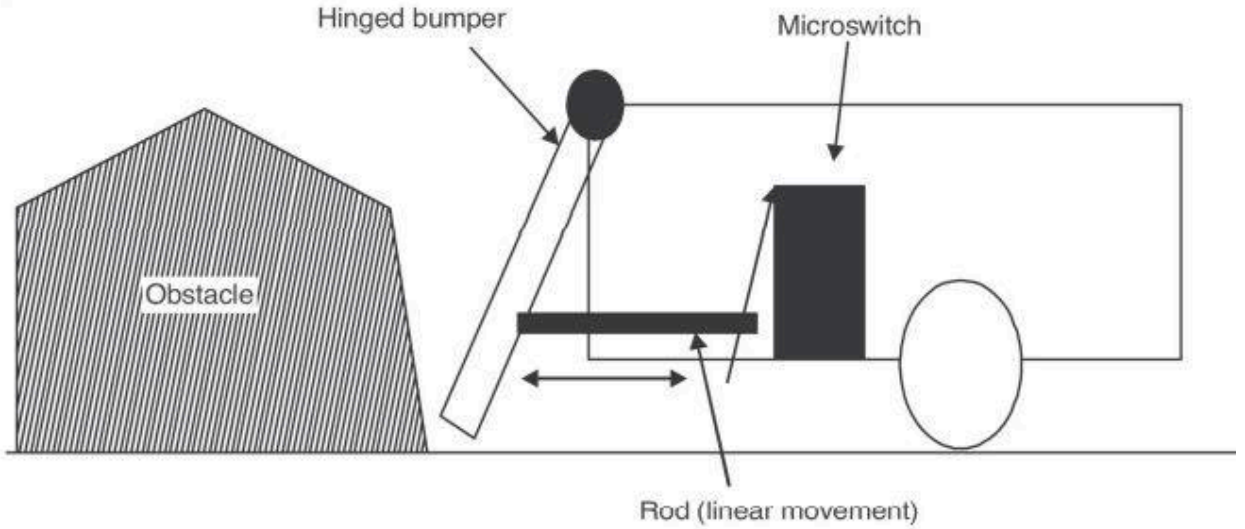



*FIGURE 14.2* Robotic Platform Employing Bumper Coupled to Touch Sensor.

# Force/Torque Sensors

- Force/torque sensors measure the amount of force and torque exerted by the mechanical hand. These sensors are normally used to measure the robotic system's forces as it performs various operations.
- Forces, which the robot uses in manipulation and assembly, are usually of great importance. The major parts of these devices are transducers, which measure force/torque.
- The interaction forces and torques, which appear during mechanical assembly operations at the robot hand level, can be measured by sensors mounted on the joints or on the manipulator wrist.
- Individual stress sensors usually respond only to force in one direction on them. However, the combination of two or more can report forces as well as torques in two or three directions.
- Strain gauges are used to make force sensors, torque sensors, and sensors that can measure both kinds of stress simultaneously.

# Force/Torque Sensors

- The simplest type of tactile sensor is a gripper that is equipped with an array of miniature micro switches. This type can only determine the presence or absence of an object at a particular point or array of points.
- A more advanced type of tactile sensor uses arrays of pressure-sensitive piezoelectric material.
- This material conducts an electrical current when stressed. The more pressure applied to the material, the more electrical current is produced. This aspect allows the sensor to perceive changes in force and pressure. The tactile sensor can also be used in force feedback applications.
- These qualities include pressure, direction of forces, temperature, vibration, and texture.

# PROXIMITY SENSORS (POSITION SENSORS)

A proximity sensor is a device that senses and indicates the presence or absence of an object without requiring physical contact. This helps the workstation controller to safely move the end effectors quickly towards that object even if its position is not precisely known. The signal from the proximity detector would give the workstation controller the warning it would need in order to slow down and avoid a collision

# MACHINE VISION SENSORS

Robot vision is a complex sensing process. It involves extracting, characterizing, and interpreting information from images in order to identify or describe objects in the environment.

The repeatability and accuracy of a vision system, the ability to produce approximately the same results when given approximately the same inputs, are its greatest virtues. It can perform simple tasks such as monitoring and inspection faster and more reliably.

A vision sensor (camera) converts the visual information to electrical signals, which are then sampled and quantized by a special computer interface electronics yielding a digital image.

Solid-state CCD image sensors have many advantages over conventional tube-type sensors as: small size, lightweight, more robust, better electrical parameters, which recommend them for robotic applications.

Virtually all-existent vision sensors are designed for television, which is not necessarily best, suited for robotic applications. Because of the reduced resolution, and robot hand obstructing the field of view, the common wisdom approach of placing camera above the working area is of questionable value for many robotic applications.

*ROBOT END EFFECTORS*

# INTRODUCTION

The function of a robot is to interact with the surroundings. The robot does this by manipulating objects and tools to fulfill a given task. The robot end effector becomes a bridge between the computer controlled arm and the world around it. Earlier industrial robots were used as stand alone machines for painting, spot welding, or pick and place work in which parts were moved from one location to another without much attention paid to how the parts were picked up and put down. Nowadays, robots are put to work in more challenging applications. The tasks the robot perform may involve assembling parts or fitting them into clamps and fixtures. These tasks place greater demands on the accuracy of the arm and the end effector. The actions of the robot and the gripper determine whether the assembly will go smoothly or the parts will get damaged in the process.

# END EFFECTOR

One of the most important areas in the design of robot systems is the design of end effectors. Most of the problems that occur in production are caused by badly designed tooling and not by faults in the robots.

In robotics, an *end effector* is a device or tool connected to the end of a robot arm. Basically, it is a tool to grip, hold, and transport objects and position them in a desired location. *End effectors* are the devices through which a robot interacts with the world around it, grasping and manipulating parts, inspecting surfaces, and working on them. As such, end effectors are among the most important elements of a robotic application and an integral component of the overall tooling, fixturing, and sensing strategy. An end effector attached to the robotic arm is shown in Figure 15.1.

An end effector is often different from a human hand. It could be a tool such as a gripper, a vacuum pump, tweezers, scalpel, blowtorch, *etc.* or just about anything that helps to do the job. A robotic end-effector is any object attached to the robot flange (wrist) that serves a function. This would include robotic grippers, robotic tool changers, robotic collision sensors, robotic rotary joint, robotic press tooling, compliance device, robotic paint gun, robotic deburring tool, robotic arc welding gun, *etc.* Robot end effectors are also known as robotic peripherals, robotic accessories, robot tools or robotic tools, *end of arm tooling (EOA)*, or end-of-arm devices. Some robots can change end-effectors, and be reprogrammed for a different set of tasks. If the robot has more than one arm, there can be more than one end-effector on the same robot, each suited for a specific task.

End-of-arm tooling is defined as the subsystem of an industrial robot system that links the mechanical option of the robot (manipulator) to the art being handled or worked on. An industrial robot is essentially a mechanical arm with a flat tool mounting plate at its end that can be moved to any spatial point within its reach. End-of-arm tooling in the form of specialized devices to pick up parts or hold tools to work on parts is physically attached to the robot's tool mounting plate to link the robot to the workpiece. A robot can become a production machine only if an end effector has been attached to its mechanical arm by means of the tool mounting plate as shown in Figure 15.2. Tool mounting plate is for the interface between end effector and the controller.

The structure of an end effector, and the nature of the programming and hardware that drives it, depends on the intended task. If a robot is designed to set a table and serve a meal, then robotic hands, more commonly called grippers, are the most functional end effectors. The same or similar gripper might be used, with greater force, as pliers or a wrench for tightening nuts or crimping wire.

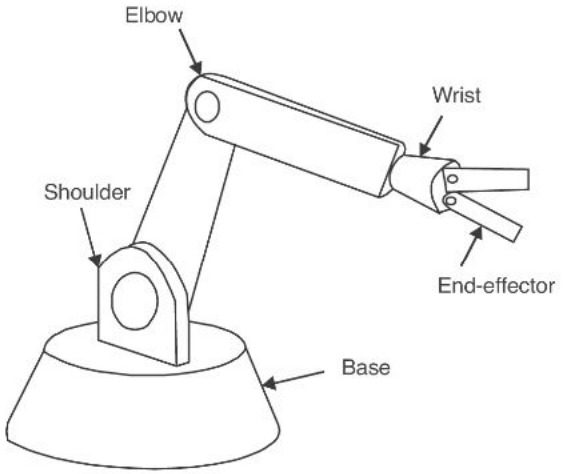



*FIGURE 15.1*

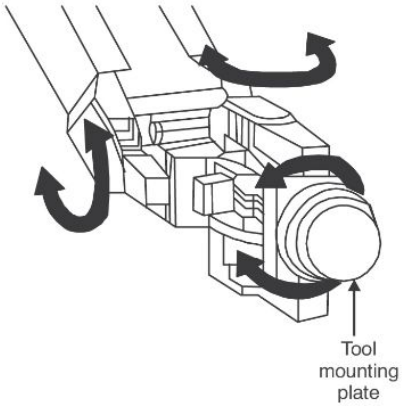



*FIGURE 15.2*

# CLASSIFICATION OF END EFFECTORS

An end effector covers a wide range of different tools and devices that can be classified into two main categories:

- Grippers
- Tools for process applications

## GRIPPERS

Grippers are end efectors that actually grip a part for transfer operations in the work envelope of the robot. A *gripper* is a component of the robot used to manipulate an object loose from the robot itself. This can be a component it needs to pick up, or the waste it is programmed to find and dispose of. Grippers include mechanical hands and also anything like hooks, magnets, and suction devices, which can be used for holding or gripping. Grippers take advantage of point-to-point control (exact path that the robot takes between what it is picking up and where it is placing it). Grippers should be designed so that it requires the minimum amount of maneuvering in order to grip the workpiece. Grippers are used in applications for material handling, machine loading, and assembling. In many grippers, the mechanism is activated through a pneumatic piston that moves the gripper's fingers. A robot gripper is shown in Figure 15.3.

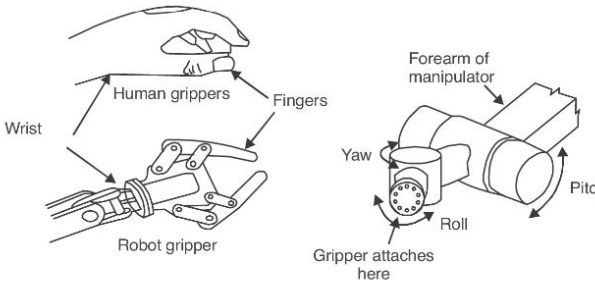



***FIGURE 15.3*** Robot Gripper.

# CLASSIFICATION OF END EFFECTORS

There are four main categories, which makes use of a gripper:

1. *No gripping*: In this situation, the workpiece is held in a jig (a specially designed purpose built holder) and the robot performs an activity on it. Jobs that use no gripping can include spot welding, flame cutting, and drilling.
2. *Coarse gripping*: In this case, the robot holds the workpiece but the gripping does not have to be precise. Jobs that use coarse gripping include handling and dipping castings, unloading furnaces, stacking boxes, or sacks.
3. *Precise gripping*: A robot holds the workpiece, which requires accurate positioning, for example, unloading and loading machine tools.
4. *Assembly*: The robot is required to assemble parts, which require accurate positioning, and some form of sensory feedback to enable the robot to monitor and correct its movements.

# SELECTION OF GRIPPER

The selection of gripper is usually made by examining the geometry of the part, its orientation, the space available, and the manufacturing treatment to be performed. External gripping is the most widely used type, where the closing force is utilized to clench the part. An internal grip makes way for unobstructed access to the outside surface of the part, which is necessary for polishing/buffing, grinding, or painting applications. The opening force of the gripper is used to hold the part. There are numerous types of grippers both in style and power source. Determining which is the best type to use is an important issue that robotics users must face. Selection of the gripper is based upon a number of factors that may need consideration:

- Source of power
- Gripping force
- Gripping style
- Weight
- Environmental capabilities
- Sensor capabilities
- Number of jaws
- Other factors

# Source of Power

According to source of power, grippers are classified as pneumatic, hydraulic, and electrical.

*(a) Pneumatic grippers* have no motors or gears. So it is simple to translate the power of a piston/cylinder system into a gripping force. Most manufacturing facilities already have compressed air, so little effort is required to bring it to a gripper in a cost efficient manner. Pneumatics put out a high amount of gripping force in a small and light package. This is critical when limited space is a factor.
There are two main types of pneumatic grippers: *angular* and *parallel*. The main difference between them is how their jaws move. The jaws of an angular gripper swing open and closed on pivots. The jaws of a parallel gripper slide open and closed in tracks. Either style can have two or three jaws.

Angular grippers cost less than parallel grippers, which require more precise machining and more expensive bearings for the jaws. They can only grasp parts from the outside. Compared with parallel grippers, angular grippers can't handle a wide range of part sizes, and they don't resist side loads as well.

# Source of Power

*(b) Hydraulic grippers* are a cost effective alternative to pneumatic grippers. The primary advantage of a hydraulic actuator is its gripping power. Hydraulic power is used when there is a need for extra gripping force. Because hydraulic fluid is not a compressible medium as is air, there is more gripping power available. Hydraulic grippers have disadvantages also. Hydraulic grippers are more costly and they are generally less accurate than pneumatic or electric grippers. Hydraulic grippers have the disadvantage of being not suited for clean room applications. Pneumatic grippers are common because they are clean, especially when compared to hydraulics. There might be more strength in a hydraulic gripper, but they are messy and much more costly to maintain.

# Source of Power

*(c)* Electricity is the other major option when looking for a source of power for robotic grippers. *Electric grippers* are a fairly new technology. They have not been cost effective until recently. Electric grippers are suitable for applications that require high speeds and for those requiring light or moderate grip forces. Electric grippers are cleaner than either pneumatic or hydraulic grippers. Electric grippers do not put out any dirt or particulate, so they are good for clean room operations.

**Gripping Force**

When considering the gripping force required, a number of factors must be considered. Not only must the mass of the object to be gripped be taken into account but also the accelerations imposed on it by the robot must be taken into account. The coefficient of friction between the gripper and the object may also be an important factor. This can often be increased by using one of the special rubber based materials that have been developed. The use of these materials can create maintenance issues, however, as they have a finite life. One other way of reducing the gripping force required is to use fingers designed for the form of the component. This reduces the flexibility of the gripper but dramatically increases the weight carrying capacity.

**Gripping Style**

The configurations of grippers include *parallel and angular styles*, and those with two or three jaws. Parallel jaws move in a motion parallel in relation to the gripper body, while angular jaws open and close around a central pivot point, moving in an arcing motion. Determining which gripper to use is dependent on the specific needs of an application.

An *angular gripper* is used when one needs to get the tooling out of the way. Angular grippers are usually the most economical. However, because of the sweeping action of an angular gripper, it is not always practical for some applications, and it is more difficult to design.

A *parallel gripper* is used for pulling a part down inside a machine because the fingers fit into small areas better. Space constraints might lead to the use of parallel over angular. Two jaw parallel grippers are the most widely used. They are the most popular type because two jaw parallel grippers are easier to design and program as there is only axis of motion.

Angular and parallel grippers are shown in Figure 15.4

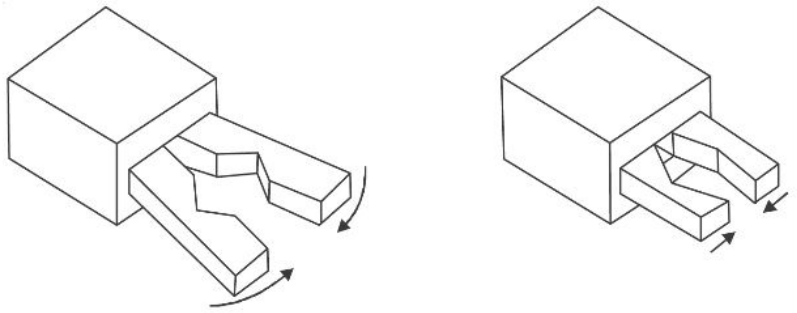



***FIGURE 15.4*** Angular and Parallel Grippers.

A gripper can also be *external or internal*. The external gripper is used to grasp the exterior surface of objects with closed fingers, whereas the internal gripper grips the internal surface of objects with open fingers. External and internal grippers are shown in Figure 15.5.

**Weight**

Industrial robots have fixed lifting capabilities and the combined weight of the gripper and gripped component may be important. Even when this weight is within the capability of the robot, it may cause an unacceptable increase in the cycle time of the operation. The distance between the robot flange and the center of mass may also be important and this should be kept to a minimum.

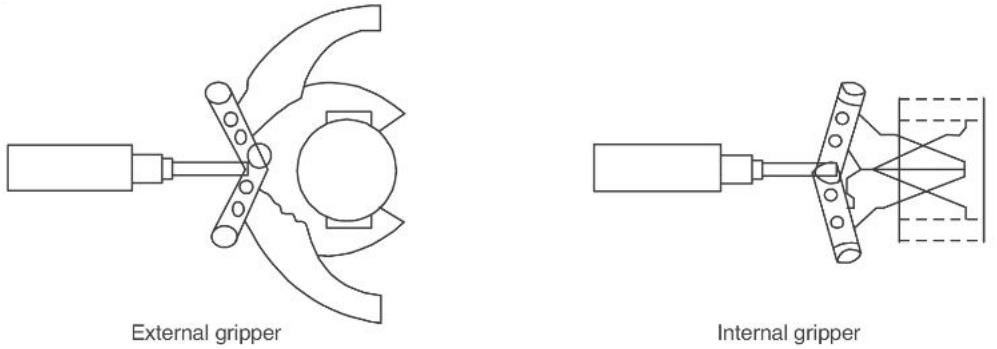



*FIGURE 15.5* External and Internal Grippers.

**Environmental Capabilities**

End effectors are often required to work in hostile environments. High temperatures, dust, or the presence of chemicals will require special materials or designs to be used.

**Sensor Capabilities**

For certain applications, some degree of sensory feedback from the gripper is necessary. This may be measurement of insertion or gripping forces or may simply be a proximity sensor to say if anything is between the jaws of the gripper. Some standard grippers are provided with feedback to show the separation of the jaws but most grippers have no feedback.

**Number of Jaws (Two or Three Jaw Gripper)**

In two jaw units, which come in both parallel and angular styles, the grasper affords two mounting spots for the fingers to hold the part. The jaws move in a simultaneous motion, opening and closing toward the central axis of the gripper body. Three jaws furnish more contact with the part to be clasped and are more accurate in centring than are two jaws.

# GRIPPING MECHANISMS

A gripper is specifically end of arm tooling that uses a mechanical mechanism and actuator to grasp a part with gripping surfaces. Fingers are designed to:

- Physically mate with the part for a good grip.
- Apply enough force to the part to prevent slipping.

**Typical mechanisms**

- Linkage actuation
- Gear and rack
- Cam
- Screw
- Rope and pulley
- Miscellaneous, e.g., bladder, diaphragm

# Robot Programming

- Robots are becoming more powerful, with more sensors, more intelligence, and cheaper components. As a result robots are moving out of controlled industrial environments into uncontrolled service environments such as homes, hospitals, and workplaces where they perform tasks ranging from delivery services to entertainment.
- It is this increase in the exposure of robots to unskilled people that requires robots to become easier to program and manage.
- The flexibility of a robot system comes from its ability to be programmed.
- A good mechanical arm can be underutilized if it is too difficult to program.
- Earlier robot programming was easy because it only required guiding the robot through the sequence of desired movements.
- To execute complete tasks of the type found in assembly, robot-programming languages had to be introduced.
- Although the introduction of robot programming languages has represented an important breakthrough in industrial robotics, currently available languages are not easy to use. The programmer must define all the movements very precisely. Programming the robot is not the same as programming the computer

# ROBOT PROGRAMMING

- A robot program can be defined as a path in space to be followed by the manipulator, combined with peripheral actions that support the work cycle.
- A robot can perform complex tasks under the control of stored programs, which can be modified at will. The process of robot programming includes teaching it the task to be performed, storing the program, executing the program, and debugging it.
- Robotic programming is similar to real-time programming in that the programs must be interrupt driven and take account of limited resources.
- Robot programs can be simple or complex.
- Simple programs will have a set of pre-programmed responses to expected events. For example, if the robot is hit, it may move in a particular direction.
- Complex robot programs may include a way to learn from past events and actions and predict what will happen. For example, a robot programmed to move left when it reaches a barrier will always move in this direction.
- However, a robot might be programmed to remember which direction has the fewest barriers and will move in that direction. This element of learning is more likely found in robot programs than real-time programs.

# ROBOT PROGRAMMING TECHNIQUES

- A number of different techniques are used to program robots.
- The principal task of robot programming is to control the motions and actions of the manipulator.
- A robot is programmed by entering the programming commands into its controller memory.
- The methods of entering the commands are:
  - Online programming
  - Lead-through programming
  - Walk-through programming
  - Offline programming
  - Task programming

# ONLINE PROGRAMMING

- Online programming takes place at the site of production itself and involves the work cell.
- Online programming systems (Refer to Figure below) uses a teach pendant to direct the robot's movement which allows trained personnel to physically lead the robot through the desired sequence of events by activating the appropriate pendant button or switch.
- Position data and functional information are "taught" to the robot, and a new program is written.
- Taught data is stored in the pendant's memory then transferred to the robot's controller.
- The teach pendant can be the sole source by which a program is established, or it may be used in conjunction with an additional programming console and/or the robot's controller.
- When using this technique of teaching or programming, the person performing the teach function can be within the robot's working envelope, with operational safeguarding devices deactivated or inoperative.
- There are a number of teach pendant types available, depending on the type of application for which they will be used.
- If the goal is simply to monitor and control a robotics unit, then a simple control box style is suitable.
- If additional capabilities, such as on the site programming are required, more sophisticated boxes should be used.

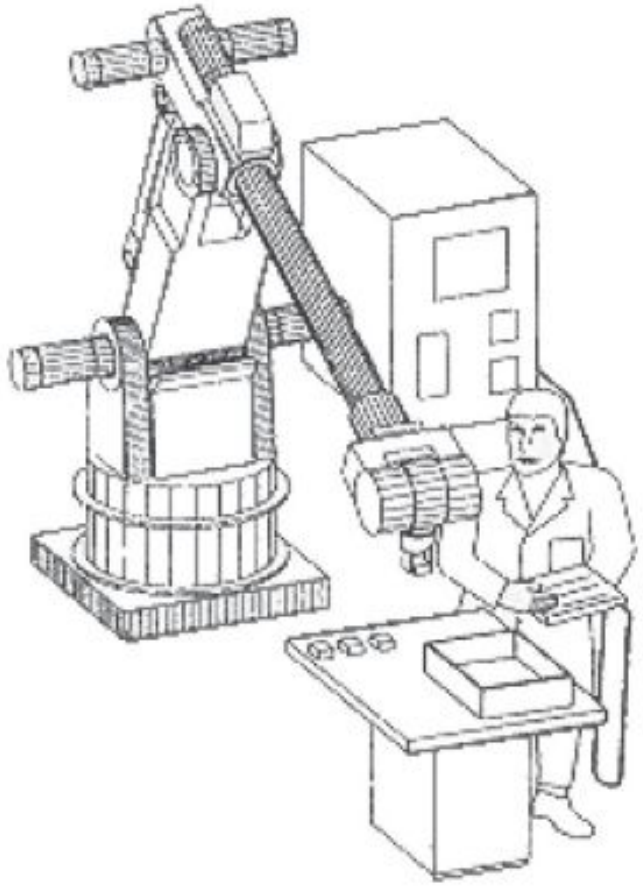


***FIGURE 16.1*** Online Programming.

Online programming is a convenient and easy method of programming when tasks are simple and revisions or adjustments can be made on the spot. However, the production line must be stopped during the programming and there are safety issues to consider, as the programmer  must work within the robot's work envelope.

# ADVANTAGES

- Easily accessible.
- The robot is programmed in concordance with the actual position of equipment and pieces.

# DISADVANTAGES

- Unavailability of robot for production during programming phase.
- Slow movement of the robot while programming.
- Time consuming process.
- Program logic and calculations are hard to program.
- Suspension of production while programming.
- Cost equivalent to production value.
- Inefficient in flexible manufacturing system.

# The Teach Pendant (Manual Control Pendant)

- The teach pendant has the following primary functions:
  - Serve as the primary point of control for initiating and monitoring operations.
  - Guide the robot or motion device, while teaching locations.
  - Support application programs.
- The teach pendant as shown in Figure 16.2 is used with a robot or motion device primarily to teach robot locations for use in application programs.
- The teach pendant is also used with custom applications that employ “teach routines,” that pause execution at specified points and allow an operator to teach or re-teach the robot locations used by the program.
- There are two styles of teach pendants:
  - the programmer’s pendant, which is designed for use while an application is being written and debugged, and
  - the operator’s pendant, which is designed for use during normal system operation

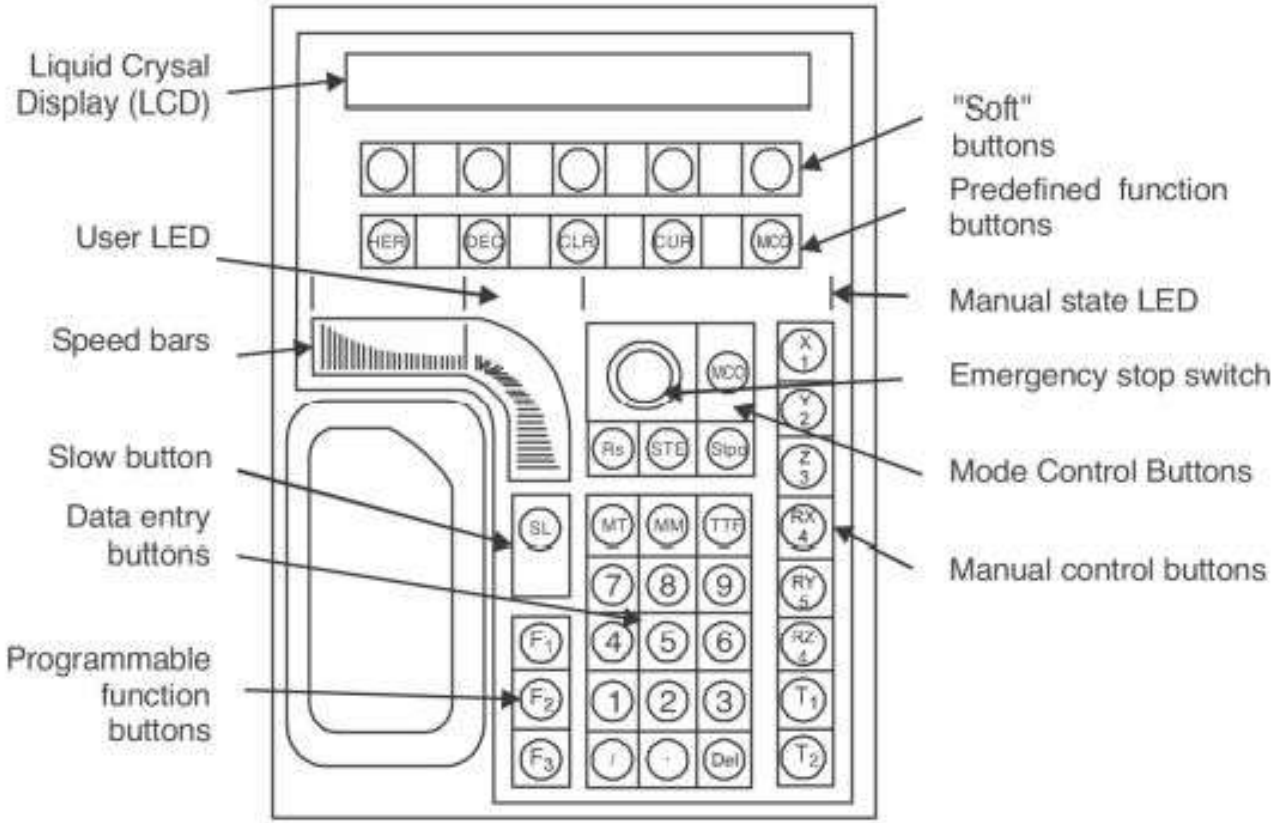



*FIGURE 16.2* Teach Pendant.

The operator's pendant has a palm-activated switch, which is connected to the remote emergency stop circuitry of the controller. Whenever this switch is released, arm power is removed from the motion device. To operate the teach pendant, the left hand is put through the opening on the left-hand side of the pendant and the left thumb is used to operate the pendant speed bars. The right hand is used for all the other function buttons.

# The major areas of the teach pendant are:

- Liquid crystal display (LCD)
- Data entry buttons: The data entry buttons are used to input data, normally in response to prompts that appear on the pendant display. The data entry buttons include YES/NO, DEL, the numeric buttons, the decimal point, and the REC/DONE button, which behaves like the Return or Enter key on a normal keyboard. In many cases, application programs have users press the REC/DONE button to signal that they have completed a task.
- Emergency stop switch: The emergency stop switch on the teach pendant immediately halts program execution and turns off arm power.
- User LED: The pendant is in background mode when the user LED is not lit and none of the predefined functions are being used. The user LED is lit whenever an application program is making use of the teach pendant.
- Mode control buttons: The mode control buttons change the state being used to move the robot, switch control between the teach pendant and the application programs, and enable arm power when necessary.
- Manual control buttons: When the teach pendant is in manual mode; these buttons select which robot

# The major areas of the teach pendant are:

- Manual state LEDs: The manual state LEDs indicates the type of manual motion that has been selected.
- Speed bars: The speed bars are used to control the robot's speed and direction. Pressing the speed bar near the outer ends will move the robot faster, while pressing the speed bar near the center will move the robot slower.
- Slow button: The slow button selects between the two different speed ranges of the speed bars.
- Predefined function buttons: The predefined function buttons have specific, system-wide functions assigned to them, like display of coordinates, clear error, etc.
- Programmable function buttons: The programmable function buttons are used in custom application programs, and their functions will vary depending upon the program being run.
- Soft buttons: The "soft" buttons have different functions depending on the application program being run, or the selection made from the predefined function buttons.

# LEADTHROUGH PROGRAMMING

- Leadthrough programming requires the operator to move the robot arm through the desired motion path during a teach procedure, thereby entering the program into the controller memory for subsequent playback.
- There are two methods of performing the leadthrough teach procedure:
  - Powered leadthrough
  - Manual leadthrough
- The difference between the two is the manner in which the manipulator is moved through the motion cycle.

## Powered leadthrough

- Powered leadthrough is commonly used as the programming method for playback robots with point-to-point control.
- It involves the use of a teach pendant (hand-held control box) which has toggle switches or contact buttons for controlling the movement of the manipulator joints.
- Using the toggle switches or buttons, the programmer drives the robot arm to the desired positions, in sequence, and records the positions into memory.
- During subsequent playback, the robot moves through the sequence of positions under its own power.

# Manual leadthrough

- Manual leadthrough is convenient for programming playback robots with continuous path control in which the continuous path is an irregular motion pattern such as in spray painting.
- This programming method requires the operator to physically grasp the end-of-arm or tools attached to the arm and manually move through the motion sequence, recording the path into memory.
- Because the robot arm itself may have significant mass and would therefore be difficult to move, a special programming device often replaces the actual robot for the teach procedure.
- The programming device has a similar joint configuration to the robot, and it is equipped with a trigger handle (or other control switch), which is activated when the operator wishes to record motions into memory.
- The motions are recorded as a series of closely spaced points.
- During playback, the path is recreated by controlling the actual robot arm through the same sequence of points.
- Powered leadthrough is the most common programming method in industry at this time. Advantages Easy to program: shop personnel can readily learn it and does not require deeper programming experience. Disadvantages Interruption in production. Teach pendant have limitations in the amount of decision making logic that can be incorporated in the program. No interface to other computer subsystems in the factory. WALK-THROUGH PROGRAMMING OR TEACHING

# ADVANTAGES

- Easy to program: shop personnel can readily learn it and does not require deeper programming experience..

# DISADVANTAGES

- Interruption in production.
- Teach pendant have limitations in the amount of decision making logic that can be incorporated in the program.
- No interface to other computer subsystems in the factory.

# WALKTHROUGH PROGRAMMING/ TEACHING

- In walk-through programming, the teacher physically moves (“walks”) the robot through the desired positions within the robot’s working envelope (Refer to Figure 16.3).
- During this time, the robot’s controller may scan and store coordinate values on a fixed-time interval basis.
- These values and other functional information are replayed in the automatic mode. This may be at a different speed than that used in the walk-through.
- This type of walk-through programming uses triggers on manual handles that move the robot.
- When the trigger is depressed the controller remembers the position. The controller would then generate the movement between these points when the program is played.
- The walk-through methods of programming require the teacher to be within the robot’s working envelope with the robot’s controller energized at least in the position sensors. This may also require that safeguarding devices be deactivated.
- A person doing the teaching has physical contact with the robot arm and actually gains control and walks the robot’s arm through the desired positions within the working envelope.
- With the walk-through method of programming, the person doing the teaching is in a potentially hazardous position because the operational safeguarding devices are deactivated or inoperative. The walk-through

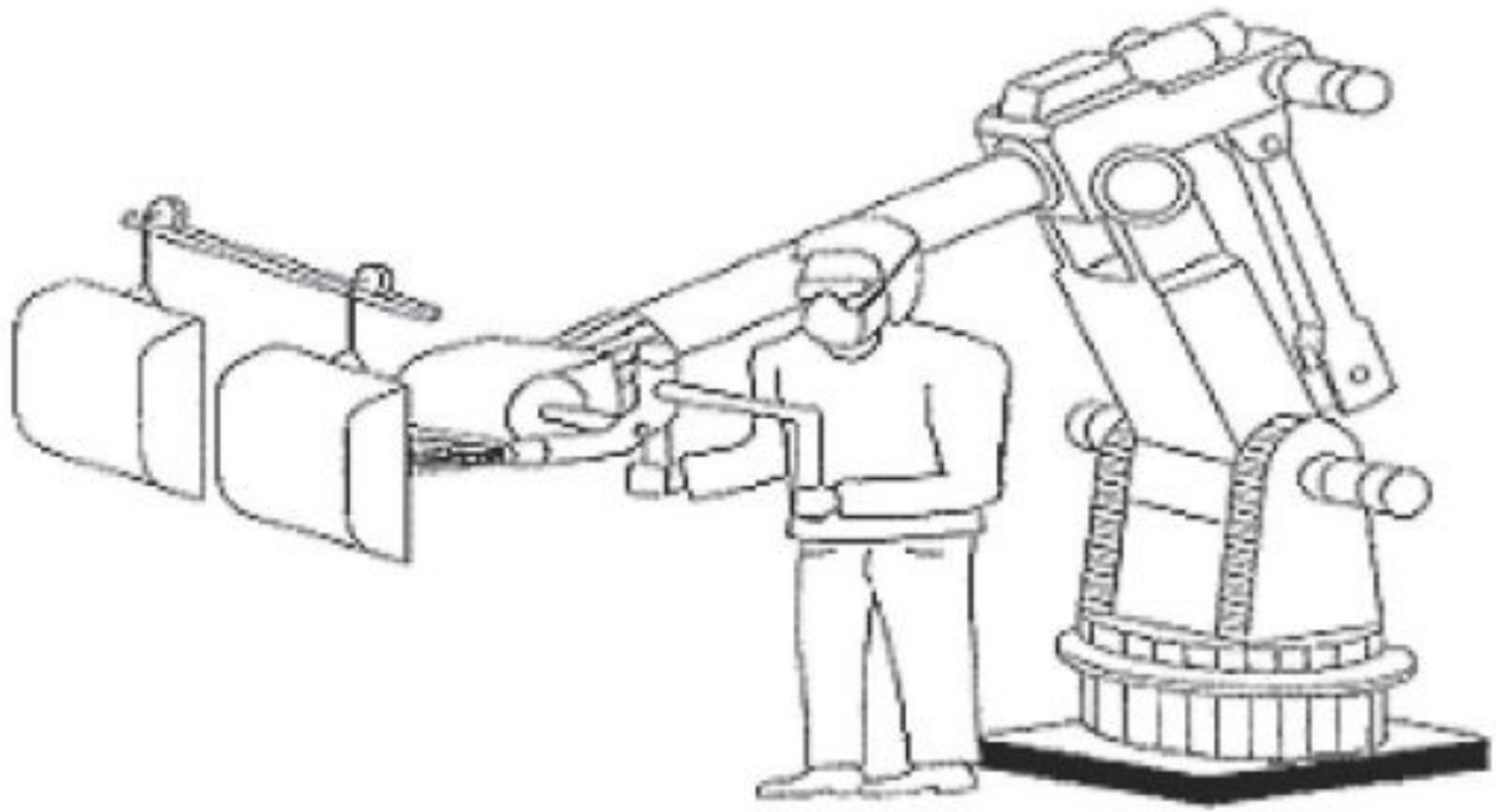


***FIGURE 16.3*** Walk-through Programming Method.

# OFFLINE PROGRAMMING

- Offline programming is accomplished on computers located away from the robot station (Refer to Figure 16.4).
- Using simulation software, data is generated and then sent to the robot's controller where it is translated into instructions.
- Additionally, the software contains modeling data, which assists selection of the best robot configuration for a particular application.
- The advantage of this programming method is that programming can be done while the robot is still in production on the preceding job, thus production time of the robot is not lost to delays in teaching the robot a new task. This ensures higher utilization of the robot.

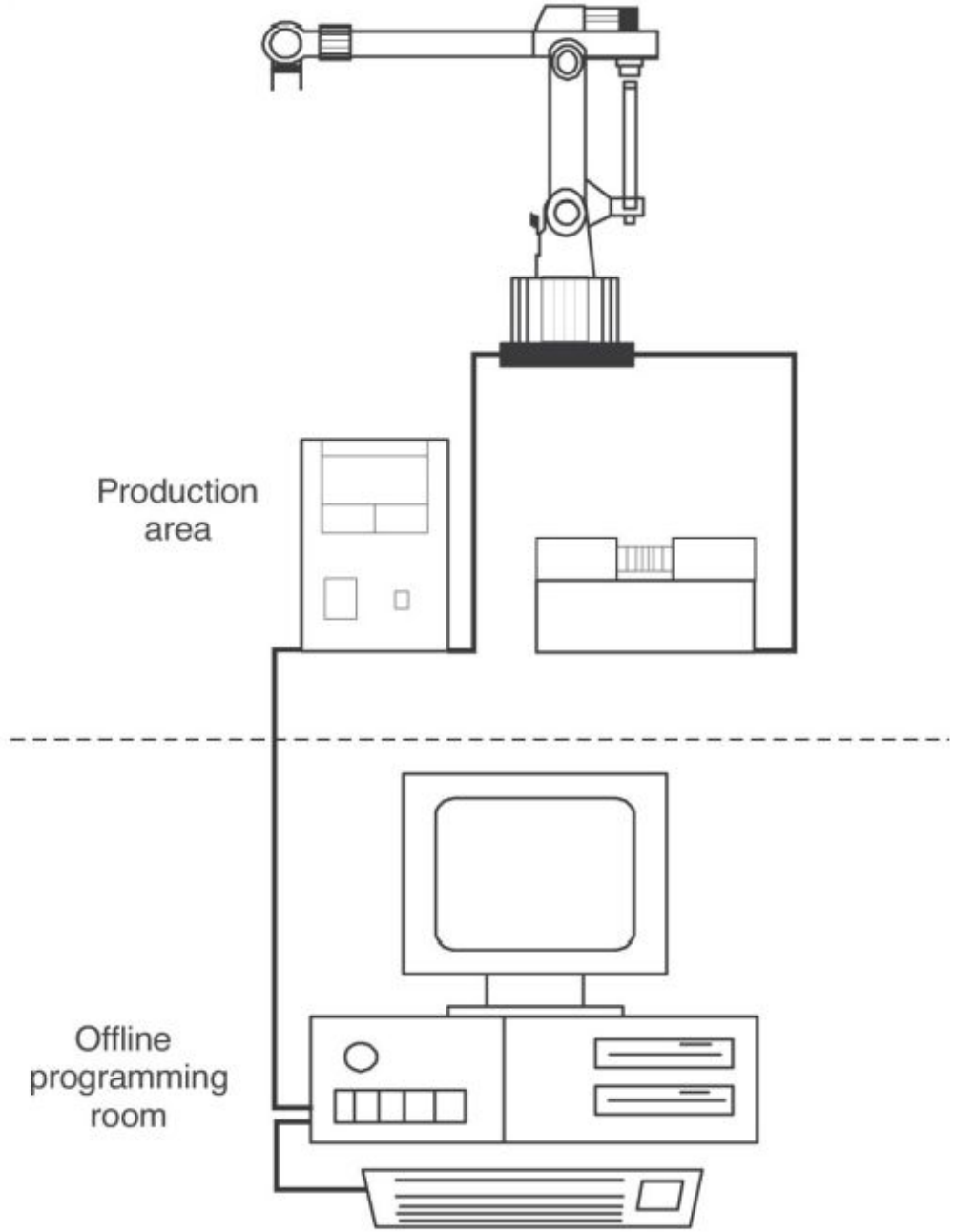



***FIGURE 16.4.*** Offline Programming.

# ADVANTAGES

- Effective programming of program logics and calculations with state of the art debugging facilities.
- Locations are built according to models and this can mean that programmers will have to fine tune programs online or utilize sensors.
- Effective programming of locations.
- Verification of program through simulation and visualization.
- Well documented through simulation model with appropriate programs.
- Reuse of existing CAD data.
- Cost independent of production.
- Production can continue while programming.
- Process support tools, for instance, selection of welding parameters.

# DISADVANTAGES

- Demands investment in an offline programming system.
- Needs extensive training.

# TASK PROGRAMMING

- Task level programming requires specifying goals for the position of objects, rather than the motions of a robot needed to achieve those goals.
- A task level specification is meant to be totally robot independent, no positions or paths that depend on robot geometry are specified by the user.
- Task level programming systems requires complete geometric models of the environment and of the robot as input.
- In the task level language, robot actions are specified by their effects on objects.
- For example, a user would specify that a job should be placed in a pallet rather than specifying the sequence of manipulator motions needed to perform the insertion.
- A task planner would transfer the task level specifications into robot level specifications.

# MOTION PROGRAMMING

- Motion programming with today's robot languages requires a combination of textual statements and leadthrough techniques.
- Accordingly, this method of programming is sometimes referred to as online/offline programming.
- The textual statements are used to describe the motion, and the leadthrough methods are used to define the position and orientation of the robot during and/or at the end of the motion.
- The leadthrough methods provide a very natural way of programming motion commands into the robot controller.
- In manual leadthrough, the operator simply moves the arm through the required path to create the program. In powered leadthrough, the operator uses a teach pendant to drive the manipulator.
- The teach pendant is equipped with a toggle switch or a pair of contact buttons for each joint.
- By activating these switches or buttons in a coordinated fashion for the various joints, the programmer moves the manipulator to the required positions in the workspace.

# MOTION PROGRAMMING

- Coordinating the individual joints with the teach pendant is sometimes an awkward way to enter motion commands to the robot.
- For example, it is difficult to coordinate the individual joints of a jointed-arm robot to drive the end-of-arm in a straight-line motion.
- Therefore, many of the robots using powered leadthrough provide two alternative methods for controlling movement of the manipulator during programming, in addition to individual joint controls.
- With these methods, the programmer can control the robot's wrist end to move in straight-line paths. The names given to these alternatives are (1) world coordinate system, and (2) tool coordinate system.
- Both systems make use of a Cartesian coordinate system. In the world coordinate system, the origin and frame of reference are defined with respect to some fixed position and alignment relative to the robot base. In the tool coordinate system, the alignment of the axis system is defined relative to the orientation of the wrist faceplate (to which the end effector is attached).
- In this way, the programmer can orient the tool in a desired way and then control the robot to make linear moves in directions parallel or perpendicular to the tool.
- The speed of the robot is controlled by means of a dial or other input device located on the teach pendant and/or the main control panel. Certain portions of the program should be performed at high speeds (e.g., moving parts over substantial distances in the workcell), while other parts of the program require low-speed operation (e.g., moves that

# OVERVIEW OF ROBOT PROGRAMMING LANGUAGES

- Virtually all robots are programmed with some kind of robot programming language.
- These programming languages are used to command the robot to move to certain locations, signal outputs, and read inputs.
- The programming language is what gives a robot its flexibility and allows a robot to react to its environment (through the use of sensors), and to make decisions. Robot programming languages fall into three basic categories:

1. Specialized robot languages
   - These languages have been developed specifically for robots.
   - The commands found in these languages are mostly motion commands with minimal logic statements available.
   - Most of the early robot languages were of this type, although many still exist today.
   - VAL is an example of this.
2. Robot library for a new general-purpose language
   - These languages were created by first creating a new general?purpose programming language and then adding robot-specific commands to it.
   - These languages are generally more capable than a specialized language, because they tend to have better logic testing capabilities.
   - KAREL language is an example of this type.

# OVERVIEW OF ROBOT PROGRAMMING LANGUAGES

3. Robot library for an existing computer language
   - These languages are developed by creating extensions to popular computer programming languages.
   - Consequently, these languages closely resemble traditional computer programming languages, and benefit from the power of existing, widely used languages.
   - Robot script is an example of this type of language.
   - Generally, all the robot programs have three basic modes of operation.
   - They are:
     - Monitor mode: In this mode, the user defines the various positions to be used in the program. These positions will be taught using the teach pendant and stored into the memory to be used in the program.
     - Editor mode: This mode is used by the user to write new programs and edit the existing ones. In this mode, the syntax checking will be done.
     - Run mode: This is the mode in which the robot is actually executing the program. The robot is actually performing the sequence of motions in the run mode.

# REQUIREMENTS FOR A STANDARD ROBOT LANGUAGE

Because of the different methods of developing robot languages, many different types of languages are available. Currently, there are no standards for robot languages, and each robot manufacturer has developed their own, each with their own syntax and data structures. Some factories have robots from multiple robot manufacturers, thus, multiple languages are running on these control systems. This requires robot programmers to be proficient in many languages, or for the robot programmers to specialize in certain languages. The result of this variety is a demand for a common language that can be used on any type of robot. To develop a new robot programming language, the deficiencies of the existing languages, as well as the requirements of a new language, need to be identified.

As robot workcells are becoming more complex, the robot has to perform more complicated moves and interact with more sensors and other peripheral devices. Robots often have to decide which part is coming down the line, and determine the proper program to run. Existing robot programming languages, especially the specialized languages, often have limited ability to use subroutines and do logic testing. Some of the early generation controllers just don’t have the memory capacity to hold such large programs.

Most robot languages require the user to compile a program before running it on the robot controller. Compiling a program converts the human?readable source code into a machine-readable form. This is usually done to increase the speed of the program execution, because the program is already in the easiest form for the computer to read. Interpreters, on the other hand, convert the human-readable code to computer-readable code on the fly, as the program executes. Usually, this results in a

# ROBOTS LANGUAGES

Some of the basic types of commands in programming languages are:

- Motion and sensing functions (e.g., MOVE, MONITOR)
- Computation functions (e.g., ADD, SORT)
- Program flow control functions (e.g., RETURN, BRANCH)

# TYPES OF ROBOT LANGUAGES

- Some of the high-level computer languages now used to program robots are: Wave, AL, ACL, AML, APT, ARCL, ZDRL, HELP, Karel, CAP 1, MML, RIPL, MCL, RAIL, RPL, ARMBASIC, Androtext, VAL, IBL, and Ladder Logic.

1] Wave: Wave was the first high-level language created for programming a robot. Standford Artificial Laboratory developed it in 1973.

2] AL: The AL (Arm Language) high-level programming language was developed at the robotics research center of Stanford University.

3] ACL: The Advanced Command Language (ACL) is a robot language that employs a user-friendly conversational command environment. Yaskaua robots use it.

# TYPES OF ROBOT LANGUAGES

4] AML: AML is the programming language used for the control of robots produced by IBM. AML is intended to provide a complete interpreted computer language along with all of the programming support typically associated with high-level programming languages.

5] APT: The Automatically Programmed Tools (APT) language is a computer language dealing with motion. Electronic Systems Laboratory of MIT developed it in 1956.

6] ARCL: ARCL (A Robot Control Language) was based on Pascal-like syntax. It was a compiled language and the developed cross-compiler required three passes before the executable code was ready to be downloaded and executed in a robot. This language has a Pascal-like syntax with sensory-control and motion control commands. An example of an ARCL-language command is MOVA (GRIP, HI, CONT, MED), which opens the gripper on the robot. ARCL emphasized sensory-based programming rather than planned trajectory motion

# TYPES OF ROBOT LANGUAGES

7] ZDRL: ZDRL (Zhe Da Robot Language) was a motion-oriented robot language. It was an interpretative system and the language was composed of system commands and program instructions. System commands were used to prepare the system for execution of user-written programs. ZDRL included 32 system commands and 37 program instructions, and it contained capabilities for program editing, file management, location-data teaching, program executing, and program debugging.

8] HELP: HELP is a high-level programming language developed for use with General Electric's Allegro assembly robot.

9] Karel: Karel, Karel 2, and Karel 3 are robot control languages used by some FANUC robot controllers.

# TYPES OF ROBOT LANGUAGES

11] MML: MML was a model-based mobile robot language that was developed at the University of California. It is a high-level offline programming language, which contains functions for high-level sensor functions, geometric model description and path planning, and others. This language contains an important concept of slow and fast functions, which architecture is essential for real-time control of robots. A slow function is executed sequentially, while a fast function is executed immediately. The second important concept is the separation of the reference and current posture, which makes precise and smooth motion control and dynamic posture correction possible.

12] RIPL: RIPL (Robot Independent Programming Language) is based on an object-oriented Robot Independent Programming Environment [RIPE]. The RIPE computing architecture consists of a hierarchical multiprocessor approach, which employs distributed general and special-purpose processors. This architecture enables the control of diverse complex subsystems in real time while co-coordinating reliable communications between them.

# TYPES OF ROBOT LANGUAGES

13] MCL: MCL is short for Manufacturing Control Language and was developed by McDonnell Douglas for the U.S. Air Force's ICAM project.

14] RAIL: RAIL is a high-level programming language developed by Automatix for use with robots and vision systems.

15] RPL: RPL is a high-level programming language developed by SRI and is used to configure automated manufacturing systems.

16] ARMBASIC: ARMBASIC is an extension of the hobbyist computer language BASIC. It was used with the Microbot Mini-Mover 5 educational robot.

17] Androtext: Androtext is a high-level computer language developed by Robotronic Corporation to make commanding a personal robot easier.

# TYPES OF ROBOT LANGUAGES

18] VAL: VAL stands for Victor's Assembly Language. VAL is a high-level programming language developed for PUMA lines of robots. The programming language is similar to BASIC. It has a complete set of vocabulary words for writing and editing robot programs.

19] IBL: IBL (Instruction Based Learning) is a method to train robots using natural language instructions. IBL uses unconstrained language with a learning robot system. A robot is equipped with a set of primitive sensory?motor procedures such as turn left or follow the road that can be regarded as an execution-level command language. The user's verbal instructions are converted into a new procedure and that procedure becomes a part of the knowledge that the robot can use to learn increasingly complex procedures. With this procedure, the robot should be capable of executing increasingly complex tasks. Because errors are always possible in human-machine communication, IBL verifies whether the learned subtask is executable. If it is not, then the user is asked for more information.

20] Ladder Logic: Ladder logic is a programming language designed to be used by electricians. It closely resembles the relay logic that appears on the inside lids or doors of dishwashers and washing machines. The